

Providing city recommendations for students migrating to the UK for studies

Developed a city recommendation system for students migrating to the UK, leveraging R for data cleaning, formatting, and analysis of multi-source datasets to identify the top five student-friendly districts based on comprehensive factor evaluation.

GitHub link: <https://github.com/Shristythapa/Providing-city-recommendation-for-students-migrating-to-uk-for-studies>

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Data discovery

International friends are looking to buy a property in Oxfordshire or Yorkshire England. Here, based on the cost of housing, internet speed and the crime rate in the area, we are to recommend to them a suitable location.

To provide useful recommendations of the location we need to consider multiple factors such as house pricing, internet speed and the level of crime. The problem states that the recommendation is for friends, they are for sure to be students, so we are going to consider the school's performance as a factor for recommendation (Find School and College Performance Data in England, n.d.).

Looking at the house price is important too when buying a property as areas with high house prices aren't ideal for students. This data is provided on the UK.gov website which provides information services and resources related to government activities, police, public services and many more (Price Paid Data, 2023).

Additionally, the internet is a necessary accommodation for a student and looking at the quality of the internet is important for better efficiency when working online (Connected Nations and Infrastructure Reports, 2023). To ensure that students buy property in a safe area, analyzing crime data is necessary (Data Downloads | data.police.uk, n.d.).

Population data is used for understanding the relative impact or occurrence of certain events or phenomena. For example, it is used to calculate crime rates according to the population of the location. Additionally, PostcodeToLOAS data is used to get additional information on the location.

The place recommendation is done from two countries in England oxfordshire, and Yorkshire and the data provided by the UK government on their website is the most valid source of data. Therefore, all the data used will be public information provided by the UK government.

Ethical and Legal Considerations

The project complied with the ethical and legal requirements. Here all data is collected for reputable websites that include sites hosted by the UK government for providing free information for anyone interested. The document is transparent about the methods used for data collection, cleaning, analysis, and visualization. Assumptions and limitations were openly discussed to provide readers with a clear understanding of the study's scope. This way it has enabled others to replicate the analysis and verify the results independently.

Data cleaning

Tidyverse library is used to make data cleaning more efficient and intuitive. It includes all packages required for data science workflow where reader and tidy are used specifically for data cleaning and additionally contains stringr package used for manipulation of strings (Avcontentteam, 2020). Lubridate package is used to work with date formats.

Figure 1

Packages to Clean Data

```
#install.packages("tidyverse")
library(tidyverse)

#install.packages("lubridate")
library(lubridate)
```

Read and clean postcode to load data.

Figure 2

Read postcode data.

```
PostcodeToLAOS <- read_csv("dataset\\Postcode to LSOA.csv")
```

Figure 3

Postcode data before cleaning

```
> #taking look at the postcodetolaos data
> glimpse(PostcodeToLAOS)
Rows: 2,632,804
Columns: 14
$ pcd7      <chr> "AB1 OAA", "AB1 OAB", "AB1 OAD", "AB1 OAE", "AB1 OAF", "AB1 OAG", "AB1 OAJ", "AB1 OAL",...
$ pcd8      <chr> "AB1 OAA", "AB1 OAB", "AB1 OAD", "AB1 OAE", "AB1 OAF", "AB1 OAG", "AB1 OAJ", "AB1 OAL",...
$ pcdd      <chr> "AB1 OAA", "AB1 OAB", "AB1 OAD", "AB1 OAE", "AB1 OAF", "AB1 OAG", "AB1 OAJ", "AB1 OAL",...
$ dointr    <dbl> 198001, 198001, 198001, 199402, 199012, 199012, 198001, 198001, 198001, 198001, 198001,...
$ doterm     <dbl> 199606, 199606, 199606, 199606, 199207, 199207, 199606, 199606, 199606, 199606, 199606,...
$ usertype   <dbl> 0, 0, 0, 0, 1, 1, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ...
$ oa11cd     <chr> "S00090303", "S00090303", "S00090399", "S00091322", "S00090299", "S00090291", "S0009039...
$ lsoa11cd   <chr> "S01006514", "S01006514", "S01006514", "S01006853", "S01006511", "S01006506", "S0100651...
$ msoa11cd   <chr> "S02001237", "S02001237", "S02001237", "S02001296", "S02001236", "S02001236", "S0200123...
$ ladcd      <chr> "S12000033", "S12000033", "S12000033", "S12000034", "S12000033", "S12000033", "S1200003...
$ lsoa11nm   <chr> "Cults, Bieldside and Milltimber West - 02", "Cults, Bieldside and Milltimber West - 02...
$ msoa11nm   <chr> "Cults, Bieldside and Milltimber Wes", "Cults, Bieldside and Milltimber Wes", "Cults, B...
$ ladnm      <chr> "Aberdeen City", "Aberdeen City", "Aberdeen City", "Aberdeenshire", "Aberdeen City", "A...
$ ladnmw     <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,...
```

The alphanumeric postcodes range in length from six to eight characters and are alphanumeric (Postal Geographies - Office for National Statistics, n.d.). Postcode area, postcode district, postcode sector, and postcode unit make up each postal code. Lsoa11cd is LSOA which

is a geographic hierarchy designed to improve the reporting of small area statistics in England.

LSOA provides a consistently sized statistical unit (North, 2023). This dataset can provide specific information on the location of entry when joined with another dataset.

Figure 4

Checking for duplicates in postcode data.

```
> #checking if duplicate rows exist
> dim(PostcodeToLAOS)
[1] 2632804      14
>
> dim(unique(PostcodeToLAOS))
[1] 2632804      14
```

Figure 5

Selecting values in postcode data.

```
#taking only required values
PostcodeToLAOS<- PostcodeToLAOS %>%
  select(pcds,lsoa11cd,ladnm)
```

Figure 6

Renaming column in postcode data.

```
#renaming columns to a meaning full name for better understading
PostcodeToLAOS <- PostcodeToLAOS %>% rename(District = ladnm)

PostcodeToLAOS <- PostcodeToLAOS %>% rename(Postcode = pcds)

PostcodeToLAOS <- PostcodeToLAOS %>% rename(LSOA = lsoa11cd)
```

Figure 7

Creating short postcode in postcode data.

```
#short post code provides information on district to perform district wise evaluation
PostcodeToLAOS <-PostcodeToLAOS %>%
  mutate(shortPostcode = str_extract(Postcode, "^^[[:space:]]+"))
```

Figure 8

Removing space in postcode.

```
# Remove spaces for consistent data
PostcodeToLAOS$Postcode <- str_remove_all(PostcodeToLAOS$Postcode, "\\s")
```

Figure 9

Clean postcode data.

```
> #taking a look at clean dataset
> glimpse(PostcodeToLAOS)
Rows: 2,632,804
Columns: 4
$ Postcode      <chr> "AB10AA", "AB10AB", "AB10AD", "AB10AE", "AB10AF", "AB10AG", "AB10AJ", "AB10...
$ LSOA          <chr> "S01006514", "S01006514", "S01006514", "S01006853", "S01006511", "S01006506...
$ District      <chr> "Aberdeen City", "Aberdeen City", "Aberdeen City", "Aberdeenshire", "Aberde...
$ shortPostcode <chr> "AB10", "AB10", "AB10", "AB10", "AB10", "AB10", "AB10", "AB10", "AB10", "AB..."
```

Figure 10

Writing clean postcode data.

```
#write clean data into a separate csv file
write_csv(PostcodeToLAOS, "clean data\\PostCodeToLAOS.csv")
```

Read and clean population data

Figure 11

Read population data.

```
population<- read_csv("dataset\\Population2011.csv")
```

Figure 12

Population data before cleaning

```
> glimpse(population)
Rows: 8,035
Columns: 2
$ Postcode <chr> "AL1 1", "AL1 2", "AL1 3", "AL1 4", "AL1 5", "AL10 0", "AL10 8", "AL10 9", "AL2 1", "AL2 2", "AL2 3...
$ Population <dbl> 5453, 6523, 4179, 9799, 10226, 9935, 10998, 14967, 9507, 6130, 7967, 5565, 8831, 4281, 5344, 4441, 9308, 65...
```

Population data simply provides us with information on the postcode and the respective population.

Figure 13

Checking for duplicates in population data.

```
#checking for duplicate data
dim(population)
dim(distinct(population))
```

Figure 14

Creating short postcode in population data.

```
#shortpostcode for year wise calculation
population <-population %>%
  mutate(shortPostcode = str_extract(Postcode, "^[^[:space:]]+"))
```


Figure 15

Removing spaces.

```
# Remove spaces for consistent data
population$Postcode <- str_remove_all(population$Postcode, "\\s")
```

Figure 16

Clean population data

```
> glimpse(population)
Rows: 8,035
Columns: 3
$ Postcode      <chr> "AL11", "AL12", "AL13", "AL14", "AL15", "AL100", "AL108", "AL109", "AL21", "AL22", "A...
$ Population    <dbl> 5453, 6523, 4179, 9799, 10226, 9935, 10998, 14967, 9507, 6130, 7967, 5565, 8831, 4281...
$ shortPostcode <chr> "AL1", "AL1", "AL1", "AL1", "AL1", "AL10", "AL10", "AL10", "AL2", "AL2", "AL2", "AL3"...
```

Figure 17

Write clean population data into CSV.

```
write_csv(population, "clean data\\populationData.csv")
```

Read and Clean housing data**Figure 18**

Looking at housing data.

```
#looking at one data file
housingData2021<- read_csv("dataset/housePricing/housepricing-2021.csv")

head(housingData2021)
```

Figure 19*Housing data before cleaning*

```
A tibble: 6 × 16
  {C6209F5F-03EF-295E-E053-6C04A8C...} `245000` `2021-05-24 00:00` `SA2 7UX` T N F `23` ...9 BRYNAERON DUNVANT SWANSEA...12
  <chr> <dbl> <dtm> <chr> <chr> <chr> <chr> <chr> <chr> <chr> <chr>
  {C6209F5F-03F0-295E-E053-6C04A8C0...} 316500 2021-05-10 00:00:00 LL55 1BE S N F GWEL... NA NORTH RO... NA CAERNARFON
  {C6209F5F-03F2-295E-E053-6C04A8C0...} 183000 2021-03-18 00:00:00 LL30 1YQ D N F 26 NA KINGSWAY NA LLANDUDNO
  {C6209F5F-03F4-295E-E053-6C04A8C0...} 155000 2021-03-26 00:00:00 CF72 9FW S N F 65 NA LONYDD G... LLANHA... PONTYCLUN
  {C6209F5F-03F5-295E-E053-6C04A8C0...} 148000 2021-03-19 00:00:00 SA4 8RP S N F 9 NA GOLWG Y ... PONTAR... SWANSEA
  {C6209F5F-03F6-295E-E053-6C04A8C0...} 136000 2021-05-21 00:00:00 CF11 7JT F N L 46 NA Ffordd J... NA CARDIFF
  {C6209F5F-03F7-295E-E053-6C04A8C0...} 260000 2021-01-22 00:00:00 SA71 4AR D N F 6 NA ASHLEIGH... NA PEMBROKE
```

Looking at the dataset there are no columns in the data set so below creating a column vector add those columns to each data set in the folder and combine them. The column names are taken from the same site where the data was downloaded (How to Access HM Land Registry Price Paid Data, 2016).

Figure 20*Read housing data.*

```
#defining folder path
folder_path <- here("dataset", "housePricing")

# Get a list of all CSV files in the folder
csv_files <- list.files(path = folder_path, pattern = "*.csv", full.names = TRUE)

#Column names
column <- c("Transaction unique identifier", "Price", "Date of Transfer", "PostCode", "Property Type ", "Old/New"
, "Duration", "PAON", "SAON", "Street", "Locality", "Town/City", "District", "Country", "Data Item ", "PDF Category Type")

#create a empty dataframe to store housePricing data
housePricingData <- data.frame()

count=0
for (file in csv_files) {
  # Read the current CSV file
  current_data <- read_csv(file)
  colnames(current_data) <- column

  # Append the current data frame to housePricing
  housePricingData <- bind_rows(housePricingData, current_data)
  count=count+1
  print(count)
}
```


Figure 24

Create a country column.

```
#Taking data of only Oxfordshire and Yorkshire countries and renaming all of Yorkshire country names to Yorkshire.
housePricingData=housePricingData %>%
  filter(Country=="OXFORDSHIRE" | Country=="EAST RIDING OF YORKSHIRE" | Country=="WEST YORKSHIRE" |
         Country=="NORTH YORKSHIRE" | Country=="SOUTH YORKSHIRE") %>%
  mutate(Country = case_when(
    endsWith(Country, "YORKSHIRE") ~ "YORKSHIRE",
    Country == "OXFORDSHIRE" ~ "OXFORDSHIRE"
  ))
```

Figure 25

Select the required data.

```
#Selecting columns relevant to our analysis.
housePricingData<- housePricingData %>%
  select(Price,`Date of Transfer`,PostCode ,District,Country)
```

Figure 26

Titled data in district house pricing

```
#change to title for consistent data
housePricingData$District<-str_to_title(housePricingData$District)
```

Figure 27

Create a short postcode in housing data.

```
#shrotpostcode for district wise calculations
housePricingData <-housePricingData %>%
  mutate(shortPostcode = str_extract(PostCode, "^[^[:space:]]+"))
```

Figure 28

Remove space in the postcode column.

```
# Remove spaces for consistent data
housePricingData$PostCode <- str_remove_all(housePricingData$PostCode, "\\s")
```

Figure 29

Change the name of the columns.

```
#For consistent naming
housePricingData <- housePricingData %>%
  rename(Postcode = PostCode)
```

Figure 30

Remove null values.

```
#removing null values
housePricingData <- housePricingData %>%
  filter(!is.na(Price) & !is.na(Postcode))
```

Figure 31

Clean housing dataset

```
> #clean dataset
> glimpse(housePricingData)
Rows: 372,333
Columns: 5
$ Price                <dbl> 145000, 180000, 140000, 140000, 92000, 140000, 5275...
$ `Date of Transfer`   <dtm> 2019-06-07, 2019-05-24, 2019-06-14, 2019-05-28, 20...
$ Postcode             <chr> "S649SN", "DN106RE", "S661WU", "S661WU", "S612HL", ...
$ District             <chr> "Doncaster", "Doncaster", "Rotherham", "Rotherham",...
$ Country              <chr> "YORKSHIRE", "YORKSHIRE", "YORKSHIRE", "YORKSHIRE",...
```

Figure 32

Writing clean housing data set into a file.

```
#write into csv file
output_path <- file.path("clean data", "housingData.csv")
write_csv(housePricingData, output_path)
```

Read and clean internet dataset

Figure 33

Read and combine Internet data.

```
#reading data add adding year column accordingly
year_2019_OX<- read_csv("dataset\\internet speed\\2019 internet\\201905_fixed_pc_performance_r01_OX.csv")
year_2019_YO<- read_csv("dataset\\internet speed\\2019 internet\\201905_fixed_pc_performance_r01_YO.csv")

internet_2019<- bind_rows(year_2019_OX, year_2019_YO) %>%
  mutate(year=2019)

year_2020_OX<-read_csv("dataset\\internet speed\\2020 internet\\202006_fixed_pc_performance_r03_OX.csv")
year_2020_YO<- read_csv("dataset\\internet speed\\2020 internet\\202006_fixed_pc_performance_r03_YO.csv")

internet_2020<- bind_rows(year_2020_OX,year_2020_YO) %>%
  mutate(year=2020)

year_2021_OX<- read_csv("dataset\\internet speed\\2021 internet\\202105_fixed_pc_performance_r01_OX.csv")
year_2021_YO<- read_csv("dataset\\internet speed\\2021 internet\\202105_fixed_pc_performance_r01_YO.csv")

internet_2021<- bind_rows(year_2021_OX,year_2021_YO) %>%
  mutate(year=2021)

year_2022_OX<- read_csv("dataset\\internet speed\\2022 internet\\202205_fixed_pc_performance_r02_OX.csv")
year_2022_YO<- read_csv("dataset\\internet speed\\2022 internet\\202205_fixed_pc_performance_r02_YO.csv")

internet_2022<- bind_rows(year_2022_OX,year_2022_YO) %>%
  mutate(year=2022)

#conbining all data set
internetData <- bind_rows(internet_2019, internet_2020, internet_2021, internet_2022)
```

Figure 34

Internet data before cleaning

```
> #looking at the dataset
> glimpse(internetData)
Rows: 1,317,767
Columns: 31
 $ postcode
 $ postcode_space
 $ postcode_area
 $ Median download speed (Mbit/s)
 $ Average download speed (Mbit/s)
 $ Minimum download speed (Mbit/s)
 $ Maximum download speed (Mbit/s)
 $ Average download speed (Mbit/s) for lines < 10Mbit/s
 $ Average download speed (Mbit/s) for lines 10<30Mbit/s
 $ Average download speed (Mbit/s) for SFBB lines
 $ Average download speed (Mbit/s) for UFBB lines
 $ Median upload speed (Mbit/s)
 $ Average upload speed (Mbit/s)
 $ Minimum upload speed (Mbit/s)
 $ Maximum upload speed (Mbit/s)
 $ Average upload speed (Mbit/s) for lines < 10Mbit/s
 $ Average upload speed (Mbit/s) for lines 10<30Mbit/s
 $ Average upload speed (Mbit/s) for SFBB lines
 $ Average upload speed (Mbit/s) for UFBB lines
 $ Number of connections < 2 Mbit/s (number of lines)
 $ Number of connections 2<5 Mbit/s (number of lines)
 $ Number of connections 5<10 Mbit/s (number of lines)
 $ Number of connections 10<30 Mbit/s (number of lines)
 $ Number of connections 30<300 Mbit/s (number of lines)
 $ Number of connections >= 300 Mbit/s (number of lines)
 $ Number of connections >= 30 Mbit/s (number of lines)
 $ Average data usage (GB) for lines < 10Mbit/s
<chr> "AB101AU", "AB101BA", "AB101BB", ...
<chr> "AB10 1AU", "AB10 1BA", "AB10 1BB...
<chr> "AB", "AB", "AB", "AB", "AB", "AB...
<dbl> 15.6, 19.7, 13.9, 13.3, 17.7, 18....
<dbl> 17.4, 18.8, 17.3, 12.9, 19.8, 17....
<dbl> 8.1, 7.8, 4.7, 8.1, 14.3, 11.5, 3...
<dbl> 72.2, 23.8, 69.1, 16.2, 69.0, 22....
<dbl> 8.7, 8.7, 6.5, 8.1, NA, NA, 3.7, ...
<dbl> 15.9, 19.8, 16.0, 14.1, 17.5, 17....
<dbl> 72.2, NA, 69.1, NA, 69.0, NA, NA,...
<dbl> NA, NA, NA, NA, NA, NA, NA, NA, N...
<dbl> 1.1, 1.1, 1.1, 1.0, 1.1, 1.1, 1.1...
<dbl> 1.9, 1.1, 2.6, 0.9, 2.0, 1.1, 1.1...
<dbl> 0.4, 0.4, 0.8, 0.7, 0.9, 0.7, 0.8...
<dbl> 20.0, 1.3, 20.0, 1.2, 20.0, 1.3, ...
<dbl> 1.0, 0.6, 1.0, 0.7, NA, NA, 0.8, ...
<dbl> 1.1, 1.1, 1.1, 1.0, 1.1, 1.1, 1.1...
<dbl> 20.0, NA, 20.0, NA, 20.0, NA, NA,...
<dbl> NA, NA, NA, NA, NA, NA, NA, NA, N...
<dbl> NA, NA, NA, NA, NA, NA, NA, NA, N...
<dbl> NA, NA, NA, NA, NA, NA, NA, NA, 6...
<dbl> 20, 20, 7, 4, 21, 23, 8, 9, 14, 5...
<dbl> NA, NA, NA, NA, NA, NA, NA, NA, N...
<dbl> NA, NA, NA, NA, NA, NA, NA, NA, N...
<dbl> NA, NA, NA, NA, NA, NA, NA, NA, N...
```

The internet data dataset encompasses details related to the internet of Yorkshire and Oxfordshire which provides valuable insights for analyzing internet quality. Moreover, after reading the data from each year, the year column is added, and all data is then combined.

Figure 35

Checking for duplicate entries on the internet.

```
> #Checking the dimension to see if the dataset contains duplicate rows
> dim(internetData)
[1] 1317767      31
>
> dim(unique(internetData))
[1] 1317767      31
> |
```

As the dimensions of the whole dataset and when we take the distinct rows are the same, we can conclude that there are no duplicate rows.

Figure 36

Taking relevant columns.

```
#Taking only the relevant columns to solve the problem.
internetData <- internetData %>%
  select("postcode", "Average download speed (Mbit/s)")
```

Figure 37

Create a country column.

```
#Adding country column according the postcode.
internetData=internetData %>%
  mutate(Country = case_when(
    startsWith(`postcode`, "OX") ~ "Oxfordshire",
    startsWith(`postcode`, "YO") ~ "Yorkshire"
  ))
```

Figure 38

Remove null values.

```
#remove all na value in country as not relevant to yorkshire or oxfordshire
internetData <- internetData[complete.cases(internetData$Country), ]
```

Figure 39

Read clean postcode data.

```
#to get the information on which district a particular post code is related to
PostcodeToLAOS <- read_csv("clean data\\PostCodeToLAOS.csv")
```


Figure 40

Clean postcode data

```
> #looking at the post code data
> glimpse(PostcodeToLAOS)
Rows: 2,632,804
Columns: 4
$ Postcode      <chr> "AB10AA", "AB10AB", "AB10AD", "AB10AE", "AB10AF", "AB10AG", "AB10AJ", "AB10AL", "AB10AN", "AB10AP", "AB10AQ", "AB10AR", "A...
$ LSOA          <chr> "S01006514", "S01006514", "S01006514", "S01006853", "S01006511", "S01006506", "S01006514", "S01006511", "S01006514", "S010...
$ District      <chr> "Aberdeen City", "Aberdeen City", "Aberdeen City", "Aberdeenshire", "Aberdeen City", "Aberdeen City", "Aberdeen City", "Ab...
$ shortPostcode <chr> "AB10", "AB10", "AB10", "AB10", "AB10", "AB10", "AB10", "AB10", "AB10", "AB10", "AB10", "AB10", "AB10", "AB10", "A...
```

Figure 41

Join postcode data to the internet.

```
#Removing duplicate postcode entries and joining the data
internetClean<-PostcodeToLAOS %>%
  group_by(Postcode) %>%
  filter(row_number()==1) %>%
  right_join(internetData, by = c("Postcode"="postcode"))
```

Figure 42

Removing null values.

```
#removing null values
internetClean <- internetData %>%
  filter(!is.na(`Average download speed (Mbit/s)`) | !is.na(Postcode) | !is.na(District) )
```

Figure 43

Clean internet data.

```
> #clean internet data
> glimpse(internetClean)
Rows: 26,519
Columns: 7
Groups: Postcode [26,519]
$ Postcode      <chr> "OX11AG", "OX11AN", "OX11AW", "OX11A...
$ LSOA          <chr> "E01028551", "E01028522", "E01028522...
$ District      <chr> "Oxford", "Oxford", "Oxford", "Oxfor...
$ shortPostcode <chr> "OX11", "OX11", "OX11", "OX11", "OX1...
$ `Average download speed (Mbit/s)` <dbl> 32.7, 50.8, 35.5, 23.8, 28.3, 56.0, ...
$ `Average upload speed (Mbit/s)`   <dbl> 6.9, 11.5, 7.2, 3.9, 4.0, 13.6, 12.8...
$ Country       <chr> "Oxfordshire", "Oxfordshire", "Oxfordsh..."
```

Figure 44

Write clean internet data into a file.

```
#write clean data into csv file
write_csv(internetClean, "clean data\\internetData.csv")
```

Read and clean Crime Data

Figure 45

Read and combine crime data.

```
# Specify the folder path
folder_path <- "dataset\\crime"

# Retrieve a list of all CSV files in the folder
csv_files <- list.files(path = folder_path, pattern = "*.csv", full.names = TRUE)

# Create an empty data frame to store the merged data
crimeData <- data.frame()

# Iterate over each CSV file, read its contents, and append it to the merged data frame
for (file in csv_files) {
  data <- read_csv(file)
  crimeData <- rbind(crimeData, data)
}
```

Figure 46*Crime data before cleaning*

```

> #taking a look at crime data
> glimpse(crimeData)
Rows: 1,914,585
Columns: 12
$ `Crime ID`      <chr> NA, "3cb03a07229df088a924cbfc809489202454e602317c938a9c4b37fd7b269b99", ...
$ Month           <chr> "2021-01", "2021-01", "2021-01", "2021-01", "2021-01", "2021-01", "2021-...
$ Reported by     <chr> "North Yorkshire Police", "North Yorkshire Police", "North Yorkshire Pol...
$ Falls within    <chr> "North Yorkshire Police", "North Yorkshire Police", "North Yorkshire Pol...
$ Longitude       <dbl> -1.817613, -2.288325, -2.538231, -2.532366, -2.509355, -2.538564, -2.533...
$ Latitude        <dbl> 53.93143, 53.59851, 54.12048, 54.14276, 54.10257, 54.12259, 54.14395, 54...
$ Location        <chr> "On or near Denton Road", "On or near Topping Street", "On or near Crowt...
$ LSOA code       <chr> "E01010694", "E01004980", "E01027558", "E01027558", "E01027558", "E01027...
$ LSOA name       <chr> "Bradford 002C", "Bury 008D", "Craven 001A", "Craven 001A", "Craven 001A...
$ Crime type      <chr> "Anti-social behaviour", "Violence and sexual offences", "Anti-social be...
$ Last outcome category <chr> NA, "Unable to prosecute suspect", NA, NA, NA, NA, "Investigation comple...
$ Context         <lgl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, ...
> |

```

Figure 47*Checking for duplicate entries in crime data.*

```

> dim(crimeData)
[1] 1914585      12
>
> dim(unique(crimeData))
[1] 1851128      12

```

As the dimensions of the whole dataset and when we take the distinct rows are different, we can conclude that there exist duplicate rows in the dataset.

Figure 48*Removing duplicate entries in crime data.*

```

#remove duplicate rows
crimeData <- distinct(crimeData)

```

Figure 49*Format date*

```
#formate date
crimeData$Month <- as.Date(paste0(crimeData$Month, "-01"), format = "%Y-%m-%d")
```

Figure 50*Rename column.*

```
#renaming column month as date
crimeData <- crimeData %>% rename(Date = Month)
```

Figure 51*Create a year column.*

```
#adding year column for year wise calculations
crimeData<- crimeData %>%
  mutate(year= year(Date))
```

Figure 52*Select necessary columns.*

```
#taking only relevant columns
crimeData <- crimeData %>%
  select(year, `Falls within`, `LSOA code`, `Crime type`)
```

Figure 53

Create a country column.

```
#create country column according to police name
crimeData<- crimeData %>%
  mutate('Country'=case_when(
    `Falls within`=="Yorkshire Police" ~ "Yorkshire",
    `Falls within`=="Oxfordshire Police" ~ "Oxfordshire"
  ))
```

Figure 54

Read clean postcode data.

```
#using postcode to laos data to get district information
PostcodeToLAOS <- read_csv("clean data\\PostCodeToLAOS.csv")
```

Figure 55

Join postcode data with crime.

```
#taking unique entries of post code in postcodetolaos
#and joining with the common column in crime data
crimeData<- PostcodeToLAOS %>%
  group_by(LSOA) %>%
  filter(row_number() == 1) %>%
  ungroup() %>%
  right_join(crimeData, by = c("LSOA"="LSOA code"))
```

Figure 56*Clean crime data*

```

> glimpse(crimeData)
Rows: 1,671,508
Columns: 8
$ Postcode      <chr> "AB113AG", "AB113AG", "AB113AG", "AB113AG", "AB113AG", "AB113AG", "AB113AG", "AB113AG", "A...
$ LSOA          <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA...
$ District      <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA...
$ shortPostcode <chr> "AB11", "AB11", "AB11", "AB11", "AB11", "AB11", "AB11", "AB11", "AB11", "AB11", "AB11", "A...
$ year          <dbl> 2021, 2021, 2021, 2021, 2021, 2021, 2021, 2021, 2021, 2021, 2021, 2021, 2021, 2021, ...
$ `Falls within` <chr> "Yorkshire Police", "Yorkshire Police", "Yorkshire Police", "Yorkshire Police", "Yorkshire...
$ `Crime type`  <chr> "Burglary", "Criminal damage and arson", "Drugs", "other theft", "Public order", "Public o...
$ Country       <chr> "Yorkshire", "Yorkshire", "Yorkshire", "Yorkshire", "Yorkshire", "Yorkshire", "Yorkshire",...

```

Figure 57*Write clean crime data into a file.*

```

#write clean data into seperate csv file
write_csv(cleanCrimeData,"clean data\\crime.csv")

```

Read and clean School Data

As stated in Secondary accountability measure file in the secondary accountability document on the website calculating Attainment 8 for academic year 2021/22, did not include results achieved from qualifications taken between January 2020 and August 2021 (Department for Education, 2023). Here data used are from the school year September 2018 to August 2019 and the school year September 2020 to 2021 in England.

Figure 58

Read school Data.

```
#read school data
school_2018To2019 <- read_csv("dataset\\school\\2018-2019\\england_ks4final.csv")
school_2021To2022 <- read_csv("dataset\\school\\2021-2022\\england_ks4final.csv")
```

Figure 59

School Data before cleaning

```
#view school data
school_2018To2019
A tibble: 5,672 x 507
```

RECTYPE	LEA	ESTAB	URN	SCHNAME	SCHNAME_AC	ADDRESS1	ADDRESS2	ADDRESS3	TOWN	PCODE	TELNUM	PCON_CODE	PCON_NAME
<dbl>	<dbl>	<dbl>	<dbl>	<chr>	<chr>	<chr>	<chr>	<chr>	<chr>	<chr>	<chr>	<chr>	<chr>
1	201	6005	100001	City of Lond...	NA	St Gile...	Barbican	NA	Lond...	EC2Y...	02078...	E14000639	Cities o...
1	201	6007	100003	City of Lond...	NA	Queen V...	NA	NA	Lond...	EC4V...	02036...	E14000639	Cities o...
4	201	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
1	202	4104	100049	Haverstock S...	NA	24 Have...	Chalk F...	NA	Lond...	NW3	02072...	E14000750	Holborn ...
1	202	4166	100050	Parliament H...	NA	Highgat...	NA	NA	Lond...	NW5	02074...	E14000750	Holborn ...
1	202	6018	100065	University C...	NA	Frogna1	Hampste...	NA	Lond...	NW3	02074...	E14000727	Hampstea...
1	202	6269	100074	North Bridge...	NA	65 Ross...	NA	NA	Lond...	NW3	02072...	E14000727	Hampstea...
1	202	6307	100076	South Hampst...	NA	3 Mares...	NA	NA	Lond...	NW3	02074...	E14000727	Hampstea...
1	202	4611	100054	The Camden S...	NA	Sandall...	NA	NA	Lond...	NW5	02074...	E14000750	Holborn ...
1	202	4688	100056	William Elli...	NA	Highgat...	NA	NA	Lond...	NW5	02072...	E14000750	Holborn ...

Figure 60

School data before cleaning

```
school_2021To2022
A tibble: 5,768 x 514
```

RECTYPE	LEA	ESTAB	URN	SCHNAME	SCHNAME_AC	ADDRESS1	ADDRESS2	ADDRESS3	TOWN	PCODE	TELNUM	PCON_CODE	PCON_NAME
<dbl>	<dbl>	<dbl>	<dbl>	<chr>	<chr>	<chr>	<chr>	<chr>	<chr>	<chr>	<chr>	<chr>	<chr>
1	201	6007	100003	City of Lond...	NA	107 Que...	NA	NA	Lond...	EC4V...	02036...	E14000639	Cities o...
1	201	6005	100001	City of Lond...	NA	St Gile...	Barbican	NA	Lond...	EC2Y...	02078...	E14000639	Cities o...
1	201	6000	100544	David Game C...	NA	31 Jewr...	London	NA	NA	EC3N...	02072...	E14000639	Cities o...
4	201	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
1	202	4285	100053	Ac1and Burgh...	NA	Burghle...	NA	NA	Lond...	NW5	02074...	E14000750	Holborn ...
2	202	7137	100092	Camden Centr...	NA	Harmood...	Camden	NA	Lond...	NW1	02079...	E14000750	Holborn ...
1	202	4611	100054	The Camden S...	NA	Sandall...	NA	NA	Lond...	NW5	02074...	E14000750	Holborn ...
1	202	6004	142408	Ecole Jeanni...	NA	43-45 B...	Bloomsb...	NA	Lond...	WC1B...	02038...	E14000750	Holborn ...
1	202	6387	100084	Fine Arts Co...	NA	Centre ...	41-43 E...	Belsize...	Lond...	NW3	02075...	E14000727	Hampstea...
1	202	6000	137333	Guildhouse S...	NA	43-45 B...	NA	NA	Lond...	WC1A...	02078...	E14000750	Holborn ...

The above dataset provides information on the school, its location and student's performance that enables analysis of the school's academic performance.

Figure 61

Adding year column.

```
#adding year column
school_2018To2019<- school_2018To2019 %>%
  mutate(year = case_when(
    sapply(strsplit(AGERANGE, "-"), `[`, 1)%in%c("9","10","11","12") ~ "2018",
    sapply(strsplit(AGERANGE, "-"), `[`, 1)%in%c("1","2","3","4","5","6","7","8") ~ "2019",
  ))
school_2021To2022<- school_2021To2022 %>%
  mutate(year = case_when(
    sapply(strsplit(AGERANGE, "-"), `[`, 1)%in%c("9","10","11","12") ~ "2021",
    sapply(strsplit(AGERANGE, "-"), `[`, 1)%in%c("1","2","3","4","5","6","7","8") ~ "2022",
  ))
```

Figure 62

Take the required data.

```
#take only necessary data from school data from both data set
school_2018To2019<- school_2018To2019 %>%
  select("SCHNAME","PCODE","ATT8SCR","year")

school_2021To2022<- school_2021To2022 %>%
  select("SCHNAME","PCODE","ATT8SCR","year")
```

Figure 63

Join two school data.

```
#combine data set of two school year
schoolData<- bind_rows(school_2021To2022,school_2018To2019)
```


Figure 64

Checking for duplicate entries.

```
> #checking for duplicate data
> dim(schoolData)
[1] 11440      4
> dim(distinct(schoolData))
[1] 10810      4
```

Figure 65

Taking only distinct values.

```
#remove duplicate rows
schoolData<- distinct(schoolData)
```

Figure 66

Creating country column.

```
#take observation related to oxfordshire and yorkshire
#add country column to dataset using postcode data
schoolData <- schoolData %>%
  filter(startswith(PCODE, "OX") | startswith(PCODE, c("YO"))) %>%
  mutate(Country = ifelse(startswith(PCODE, "OX"), "OXFORDSHIRE", "YORKSHIRE"))
```

Figure 67

Format postcode column

```
# Remove spaces for consistent data
schoolData$PCODE <- str_remove_all(schoolData$PCODE, "\\s")
```

Figure 68

Rename columns.

```
#renaming column for better understanding
schoolData<-schoolData %>%
  rename("Attainment 8 score"="ATT8SCR", "School Name"=SCHNAME)
```

Figure 69

Removing null values.

```
#removing null values
schoolData <- schoolData %>%
  filter(!is.na(`Attainment 8 score`) | !is.na(PCODE))
```

Figure 70

Change data type

```
#change attainment 8 score to numeric data
schoolData$`Attainment 8 score`<- as.numeric(schoolData$`Attainment 8 score`)
```

Figure 71

Read clean postcode data.

```
#postcode data to get data on district of school
PostcodeToLAOS <- read_csv("clean data\\PostCodeToLAOS.csv")
```

Figure 72

Join postcode data with school.

```
#join school data with postcode data to get data on district
schoolClean<-PostcodeToLAOS %>%
  group_by(Postcode) %>%
  filter(row_number() == 1) %>%
  ungroup() %>%
  select(Postcode, shortPostcode, District) %>%
  right_join(schoolData, by = c("Postcode"="PCODE"))
```

Figure 73

Remove null values.

```
#remove rows with null values in district
schoolClean <- schoolClean[complete.cases(schoolClean$District), ]
```

Figure 74

Clean school data

```
> #clean school data
> glimpse(schoolClean)
Rows: 270
Columns: 7
$ Postcode      <chr> "OX12AR", "OX12AR", "OX12EP", "OX12EP", "OX14HT", "OX14HT", ...
$ shortPostcode <chr> "OX12", "OX12", "OX12", "OX12", "OX14", "OX14", "OX10", "OX1...
$ District      <chr> "Oxford", "Oxford", "Oxford", "Oxford", "Oxford", "Oxford", ...
$ `School Name` <chr> "Cherwell College Oxford", "Cherwell College Oxford", "Carfa...
$ `Attainment 8 score` <dbl> 2.7, NA, 6.7, NA, 15.1, 14.0, 55.9, 53.2, 40.7, 39.1, 41.0, ...
$ Year          <chr> "2018-2019", "2018-2019", "2018-2019", "2018-2019", "2018-20...
$ Country       <chr> "OXFORDSHIRE", "OXFORDSHIRE", "OXFORDSHIRE", "OXFORDSHIRE", ...
```

Figure 75

Write clean school data into csv file.

```
#write clean school data to csv file
write_csv(schoolClean,"clean data\\schoolData.csv")
```

Exploratory data analysis

Figure 76

Packages for EDA

```
#install.packages("tidyverse")
library(tidyverse)

#install.packages("RColorBrewer")
library(RColorBrewer)

#install.packages("ggradar")
library(ggradar)

#install.packages("gridExtra")
library(gridExtra)
```

Figure 77

Display numeric values.

```
#sets the threshold at which scientific notation is used to display numeric values
options(scipen=1000)
```

To prepare the data for visualization, tidy verse's dplyr package is used. It has a pipe operator for arranging processes in a certain order, and its ggplot2 package is used to produce visuals. To add a colour theme to charts, used the library RColorBrewer (Alboukadel, 2019). Ggradar packages is used to create radar chart and gridExtra is used to place plots side by side (Auguié, 2019). The 'options(scipen=1000)' is used to display numerical values in scientific notation.

House Pricing

Figure 78

Read clean housing data.

```
housePricing <- read_csv("clean data/housingData.csv")
```

Box plot comparing house prices of districts in Oxfordshire and Yorkshire year 2022.

Figure 79

Sampling housing data

```
# Filter the data for each country and select 5 random districts
district_yorkshire <- housePricing %>%
  filter(Country == "YORKSHIRE") %>%
  distinct(District) %>%
  sample_n(size = 5, replace = FALSE) %>%
  select(District)

district_oxfordshire <- housePricing %>%
  filter(Country == "OXFORDSHIRE") %>%
  distinct(District) %>%
  sample_n(size = 5, replace = FALSE) %>%
  select(District)
```

Before visualization, 5 random distinct districts are taken from each country Oxfordshire and Yorkshire are taken as samples.

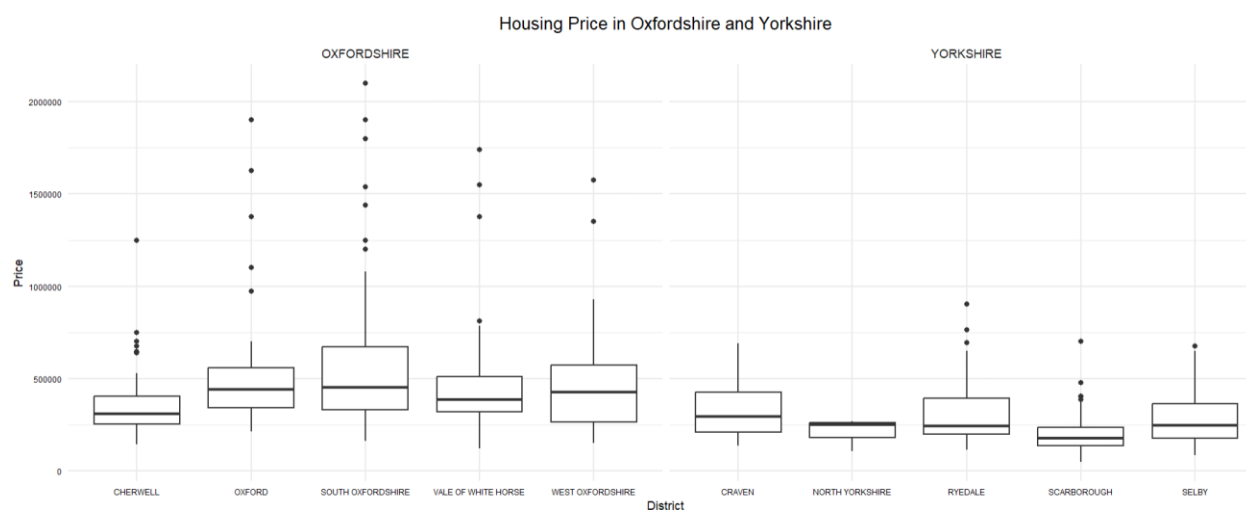
Figure 80

Code to create a box plot of house prices.

```
# Filter the data based on the selected districts and plot the boxplot
housePricing %>%
  filter((District %in% district_yorkshire$District | District %in% district_oxfordshire$District) & year(`Date of Transfer`) == 2022) %>%
  sample_n(size = 500, replace = FALSE) %>%
  ggplot(aes(x = District, y = Price)) +
  geom_boxplot() +
  facet_wrap(~ Country, scales = "free_x") +
  labs(title = "Housing Price in Oxfordshire and Yorkshire", x = "District", y = "Price") +
  theme_minimal() +
  theme(
    plot.title = element_text(hjust = 0.5),
    axis.text = element_text(color = "black", size = 6),
    axis.title = element_text(color = "black", size = 9),
    legend.title = element_blank(),
  )
```

Figure 81

Box plot of house price



The median house pricing in both Yorkshire and Oxfordshire is similar. However, Oxfordshire exhibits significant outliers. These outliers have a considerable impact on the mean, causing it to be higher compared to Yorkshire. This can be further proved below.

Figure 82

Calculate the mean of house prices.

```
housePricing %>%
  filter(Country=="OXFORDSHIRE"& year(`Date of Transfer`)==2022) %>%
  summarise(m_Oxfordshire=mean(Price))
```

```
housePricing %>%
  filter(Country == "YORKSHIRE"& year(`Date of Transfer`)==2022) %>%
  summarise(m_Yorkshire=mean(Price))
```

m_Oxfordshire	m_Yorkshire
<u>537930.</u>	<u>247385.</u>

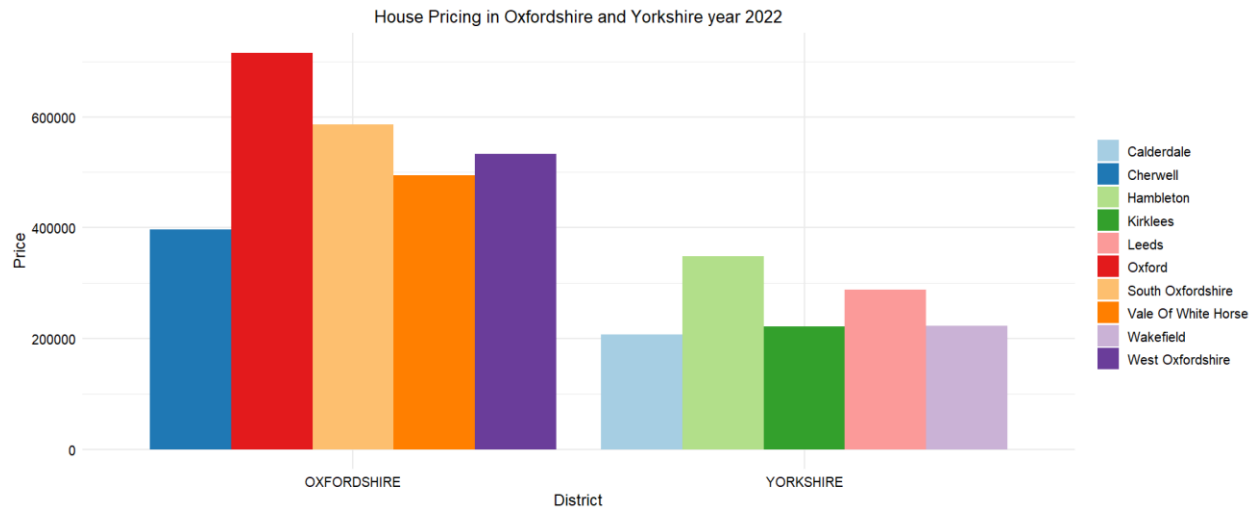
Side-by-side bar plot of Oxfordshire and Yorkshire comparing house pricing based on districts.

Figure 83

Code to create a bar graph of house prices.

```
#bargraph of house prices in districts

housePricing%>%
  filter((District %in% district_yorkshire$District | District %in% district_oxfordshire$District)&year(`Date of Transfer`)==2022) %>%
  group_by(District) %>%
  mutate(mean_housePrice=mean(Price)) %>%
  ggplot(aes(x = Country, y = mean_housePrice, fill = District)) +
  geom_bar(stat = "identity", position = "dodge") +
  labs(x = "District", y = "Price", title = "House Pricing in Oxfordshire and Yorkshire year 2022") +
  theme_minimal() +
  scale_fill_brewer(palette = "Paired")+
  theme(plot.title = element_text(hjust = 0.5),
        axis.text = element_text(color = "black", size = 10),
        axis.title = element_text(color = "black", size = 12),
        legend.title = element_blank(),
        legend.text = element_text(color = "black", size = 10))
```

Figure 84*Bar graph of house price*

In the bar chart that shows average house prices of most districts in Oxfordshire are higher than 400000 with Oxford reaching above 6000, and prices in the district in Yorkshire are less than 400000 with Calderdale having house prices of around 20000. This concludes that the prices of houses in Oxfordshire are higher than in Yorkshire.

Line graph comparing the house pricing in Oxfordshire and Yorkshire over the last 4 years.

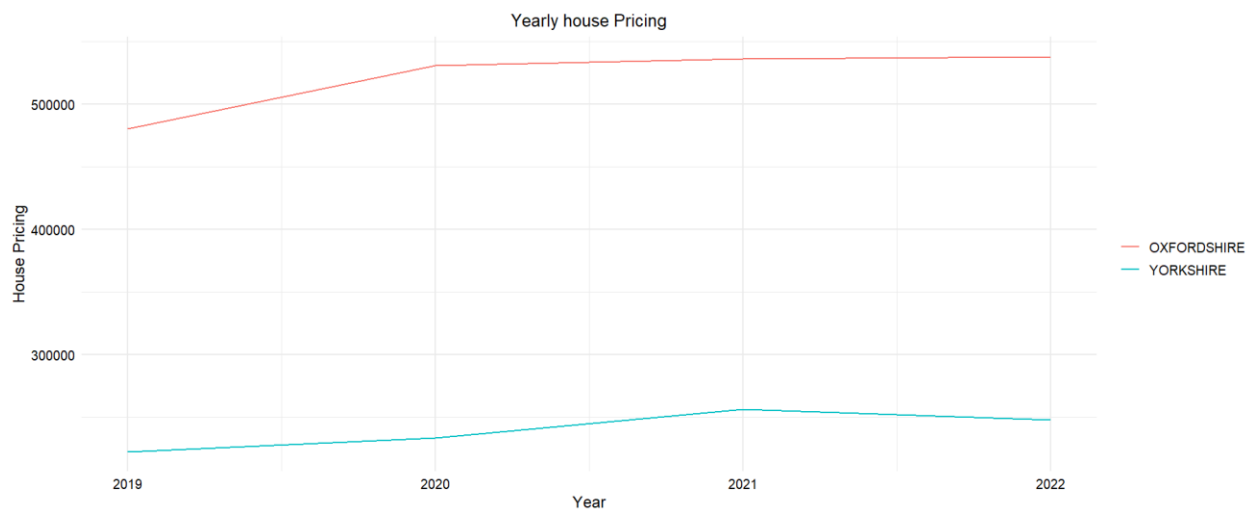
Figure 85

Code to create a line graph of house prices.

```
housePricing %>%
  group_by(Country, year(housePricing$`Date of Transfer`)) %>%
  mutate(meanPrice=mean(Price)) %>%
  ggplot(aes(x=year(`Date of Transfer`), y=meanPrice, colour= Country)) +
  geom_line()+
  theme_minimal()+
  scale_fill_brewer("PuOr")+
  labs(title = "Yearly house Pricing", x="Year", y="House Pricing")+
  theme(plot.title = element_text(hjust = 0.5),
        axis.text = element_text(color = "black", size = 10),
        axis.title = element_text(color = "black", size = 12),
        legend.title = element_blank(),
        legend.text = element_text(color = "black", size = 10))
```

Figure 86

Line graph of house price



The above chart shows the average house pricing in Oxfordshire is much higher than in Yorkshire when looking at the line graph. However, it is important to note that the higher average house price in Oxfordshire are likely to be influenced by the presence of outliers, as seen in the box plot. Moreover, the trends in house pricing over time differ as in Yorkshire, the house prices seem to have reached a plateau from the year 2021, suggesting a period of relative stability in the market. On the other hand, in Oxfordshire, house prices have shown a consistent incremental increase until the year 2020. However, after 2020, the prices appear to have remained relatively stable, forming a straight line.

Internet Data

Figure 87

Read clean Internet data.

```
internet<- read_csv("clean data\\internetData.csv")
```

Figure 88

Sample Internet data

```
#selecting 5 district from yorkshire randomly
district_yorkshire=internet %>%
  filter(Country=="Yorkshire") %>%
  distinct(District)%>%
  sample_n(size = 5, replace = FALSE) %>%
  select(District)

#selecting 5 districts from oxfordshire randomly
district_oxfordshire=internet %>%
  filter(Country=="Oxfordshire") %>%
  distinct(District)%>%
  sample_n(size = 5, replace = FALSE) %>%
  select(District)
```

Above before visualization take 5 districts from each country randomly for visualization.

Box plot visualizing average download speed of district in Oxfordshire and Yorkshire

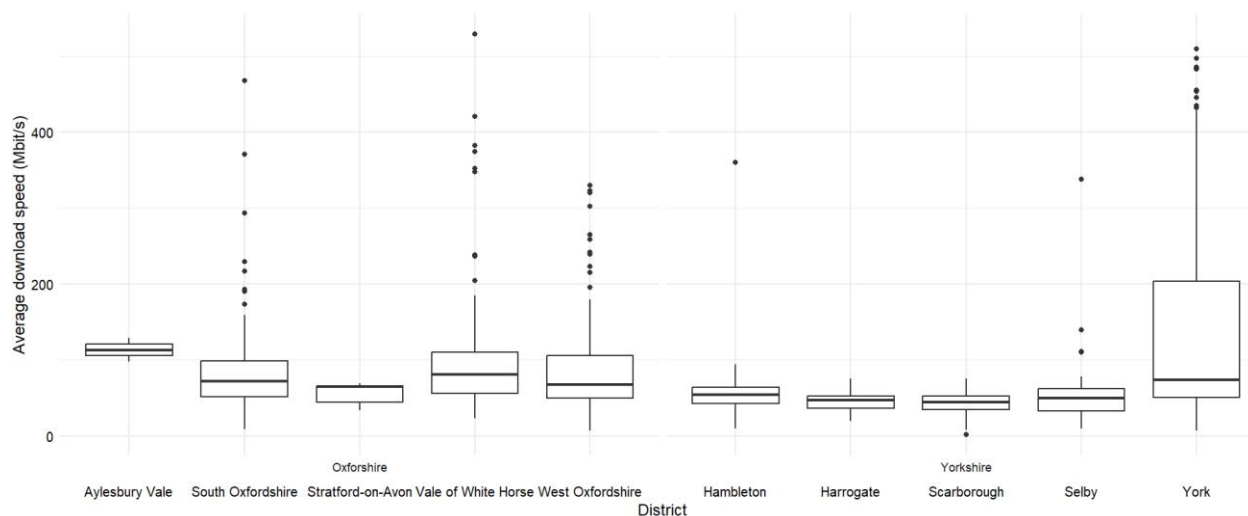
Figure 89

Code to create a box plot of average download speed.

```
#Average download speed boxplot
internet %>%
  filter(( District %in% district_yorkshire$District| District %in% district_oxfordshire$District)
    &!is.na(`Average download speed (Mbit/s)`)) %>%
  sample_n(size=900,replace = FALSE) %>%
  ggplot(aes(x = District, y = `Average download speed (Mbit/s)`)) +
  geom_boxplot() +
  facet_wrap(~ Country, nrow = 1, scales = "free_x", strip.position = "bottom") +
  theme_minimal() +
  theme(plot.title = element_text(hjust = 0.5),
    axis.text = element_text(color = "black", size = 10),
    axis.title = element_text(color = "black", size = 12),
    legend.title = element_blank(),
    legend.text = element_text(color = "black", size = 10))
```

Figure 90

Box plot of average download speed



Here in the side-by-side boxplot above there are 5 districts from each country selected randomly. We can see that the median download speeds in most of the districts are similar.

However, there are outliers in some districts in Oxfordshire and Yorkshire which means that some places in these districts have higher average download speed than others.

Bar chart of average download speed in the district in Oxfordshire

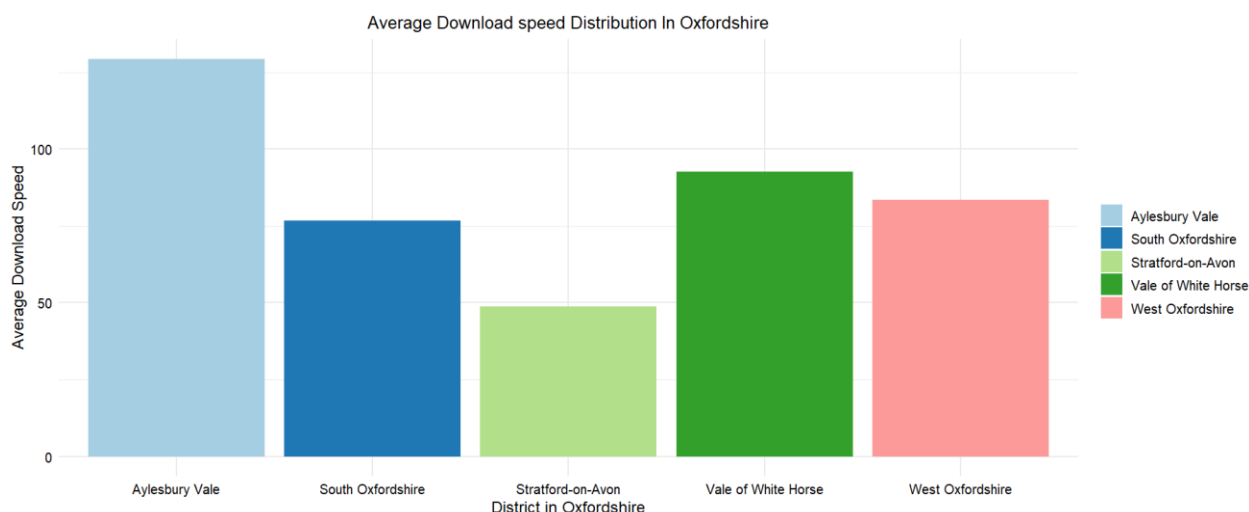
Figure 91

Code to create a bar graph of the average download speed of Oxfordshire.

```
#barchart of average download speed of district in oxfordshire
internet %>%
  filter( District %in% district_oxfordshire$District & is.na(`Average download speed (Mbit/s)`)==FALSE) %>%
  group_by(District) %>%
  mutate(ads=mean(`Average download speed (Mbit/s)`)) %>%
  filter(row_number() == 1) %>%
  ggplot(aes(x = District, y = ads,fill=District)) +
  geom_bar(stat = "identity")+
  scale_fill_brewer(palette = "Paired")+
  labs(title = "Average Download speed Distribution In Oxfordshire",x="District in Oxfordshire", y= "Average Download Speed") +
  theme_minimal() +
  theme(plot.title = element_text(hjust = 0.5),
        axis.text = element_text(color = "black", size = 10),
        axis.title = element_text(color = "black", size = 12),
        legend.title = element_blank(),
        legend.text = element_text(color = "black", size = 10))
```

Figure 92

Bar graph of average download speed in Oxfordshire



The chart above provides information on the mean of the average download speed in different districts in Oxfordshire. Here we can see that most districts have a mean Average

download speed of below 100 except for west Berkshire which has a download speed somewhat higher than other districts i.e., above 100 mits/s download speed.

Bar chart of average download speed in the district in Yorkshire

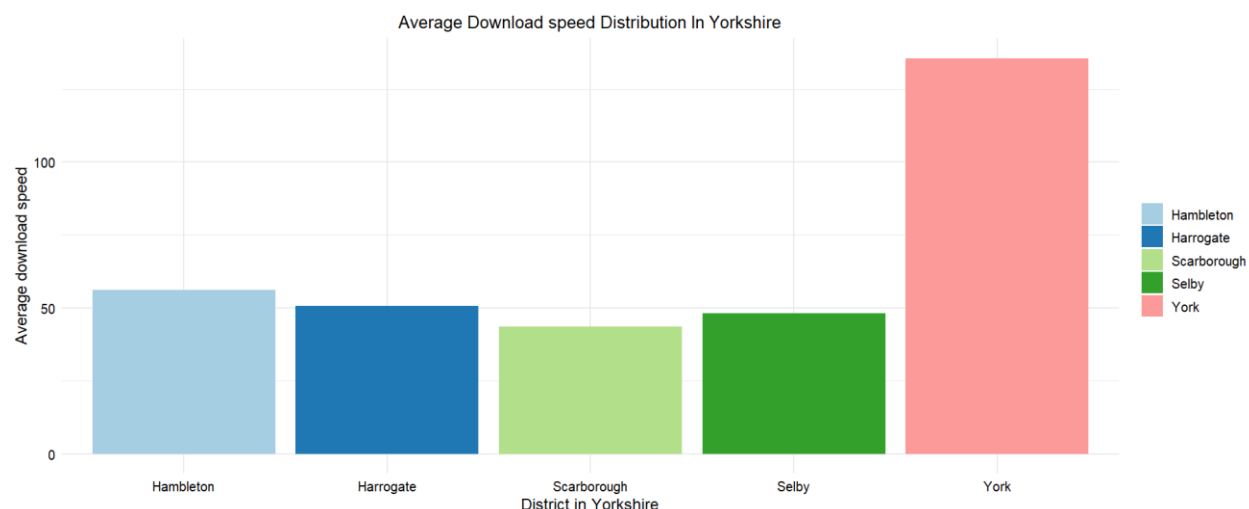
Figure 93

Code to create a bar graph of Yorkshire.

```
#barchart of average download speed of district in yorkshire
internet %>%
  filter( District %in% district_yorkshire$District & is.na(`Average download speed (Mbit/s)`)==FALSE) %>%
  group_by(District) %>%
  mutate(ads=mean(`Average download speed (Mbit/s)`)) %>%
  filter(row_number() == 1) %>%
  ggplot(aes(x = District, y = ads,fill=District)) +
  geom_bar(stat = "identity")+
  scale_fill_brewer(palette = "Paired")+
  labs(title = "Average Download speed Distribution In Yorkshire",x="District in Yorkshire",y="Average download speed") +
  theme_minimal() +
  theme(plot.title = element_text(hjust = 0.5),
        axis.text = element_text(color = "black", size = 10),
        axis.title = element_text(color = "black", size = 12),
        legend.title = element_blank(),
        legend.text = element_text(color = "black", size = 10))
```

Figure 94

Bar graph of Yorkshire



The chart above provides information on the mean of the average download speed in different districts in Yorkshire. Here we can see that most districts have a Average download speed of around 50 Mitbs/s except for the York district which has an average download speed of above 100.

The chart of Yorkshire is like Oxfordshire as one district has a higher average download than the other but in Oxfordshire, there is a higher difference in average download speed than in Yorkshire where the average download speed in one district is nearly double.

School Data

Figure 95

Read clean school Data.

```
school <- read_csv("clean data\\schoolData.csv")
```

Figure 96

Sample school data

```
#selecting 5 random district from each country
districtInYorkshire<- school %>%
  filter(Country="Yorkshire") %>%
  distinct(District) %>%
  sample_n(size=5,replace=FALSE) %>%
  select(District)

districtInOxfordine<- school %>%
  distinct(District) %>%
  sample_n(size=5,replace=FALSE) %>%
  select(District)
```

Above taking sample data i.e., 5 district data only from each country for visualization.

Boxplot comparing Attainment 8 score of Districts in Oxfordshire and Yorkshire.

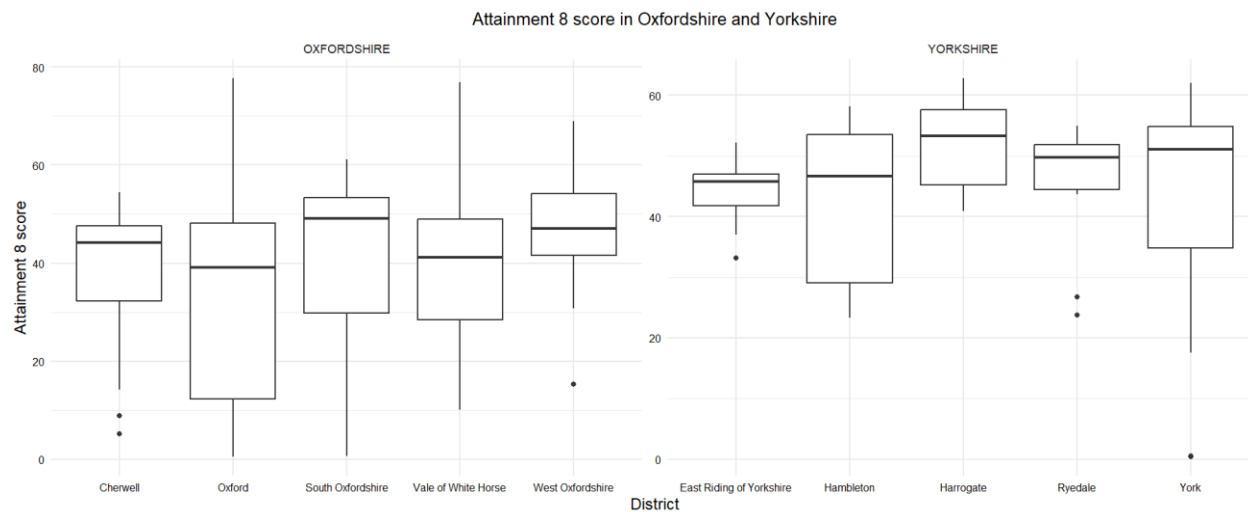
Figure 97

Code to create a box plot of average attainment 8 score.

```
#boxplot of districts in oxfordshire and yorkshire showing the average attainment rate
school %>%
  filter(District %in% districtInOxfordshire$District | District %in% districtInYorkshire$District) %>%
  na.omit() %>% # Remove rows with NA values in the ATT8SCR column
  ggplot(aes(x = District, y = ATT8SCR)) +
  geom_boxplot() +
  facet_wrap(~ Country, scales = "free") + # Replace Country with TOWN or District as per your data
  labs(title = "Average ATT8SCR by District", x = "District", y = "Average ATT8SCR") +
  theme_minimal() +
  theme(plot.title = element_text(hjust = 0.5),
        axis.text = element_text(color = "black", size = 10),
        axis.title = element_text(color = "black", size = 12),
        legend.title = element_blank(),
        legend.text = element_text(color = "black", size = 10))
```

Figure 98

Box plot of average attainment 8 score



Average attainment score in Oxfordshire contains school with a wide range of values in most of the sample districts whereas in Yorkshire there are fewer schools with low scores. The median scores of districts in Oxfordshire at 40-60 whereas in Yorkshire all the sample districts have a median attainment 8 score of above 40. Overall, we can conclude that districts in

Oxfordshire have schools scoring a wide range of different attainment 8 scores but in Yorkshire most schools have students scoring higher attainment 8 scores.

Bar graph showing average attainment of 8 scores across districts in Yorkshire of two different school years.

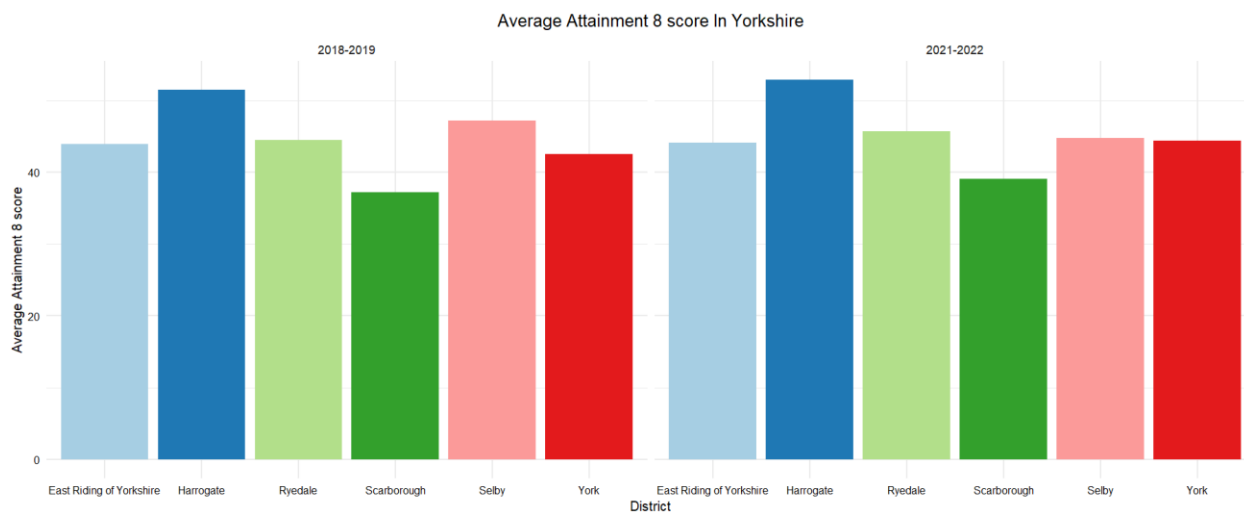
Figure 99

Code to create a bar graph of average attainment 8 scores in Yorkshire.

```
#bargraph of average attainment score in yorkshire 2018 to 2019 vs 2021 vs 2022 across
school %>%
  filter(District %in% districtInYorkshire$District) %>%
  group_by(Year, District) %>%
  mutate(ads = mean(`Attainment 8 score`, na.rm = TRUE)) %>%
  filter(row_number() == 1) %>%
  ggplot(aes(x = factor(District), y = ads, fill = factor(District))) +
  geom_bar(stat = "identity") +
  scale_fill_brewer(palette = "Paired") +
  labs(title = "Average Attainment 8 score In Yorkshire", x = "District", y = "Average Attainment 8 score") +
  facet_wrap(~Year, ncol = 2, scales = "free_x") +
  theme_minimal() +
  theme(plot.title = element_text(hjust = 0.5),
        axis.text = element_text(color = "black", size = 8),
        axis.title = element_text(color = "black", size = 10),
        legend.position = "none")
```

Figure 100

Boxplot of average attainment 8 scores in Yorkshire



The above chart shows a side-by-side bar graph of the average attainment of 8 cores of Yorkshire in 2018-2019 and 2021-2022. From above chart shows that there is very little change in the average attainment of 8 scores of all the schools in the districts of Yorkshire. One major difference can be seen where the average attainment 8 score of Seby and York are equal in the academic year 2021-2022 whereas in the year 2018-2019, York's score was lower.

Bar graph showing average attainment 8 scores across districts in Oxfordshire of two different school years.

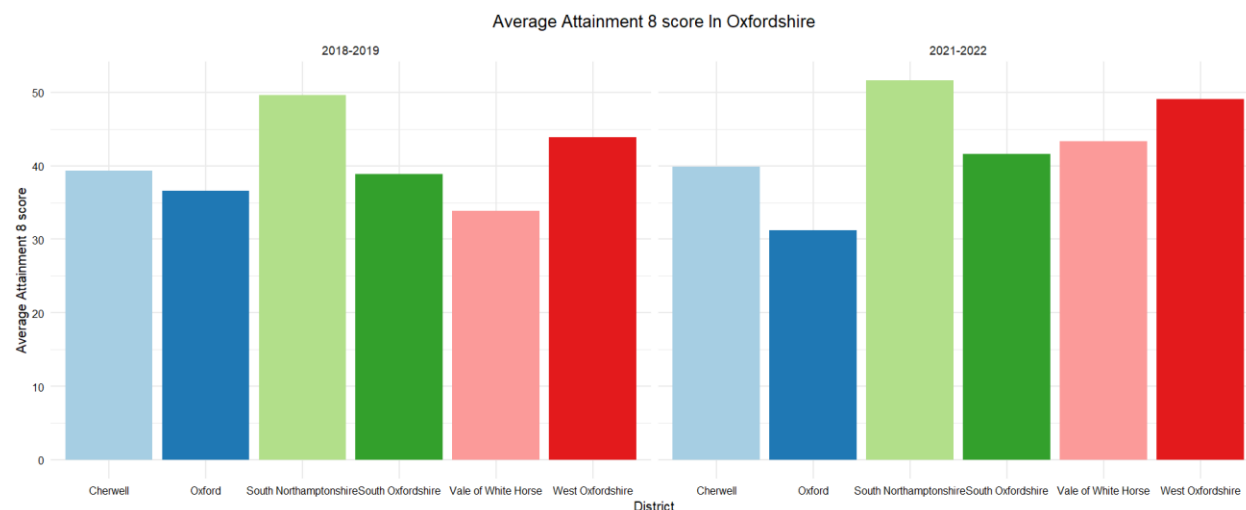
Figure 101

Code to create a bar graph of average attainment 8 scores in Oxfordshire.

```
#bargraph of average attainment score in oxfordshire 2018 to 2019 vs 2021 vs 2022 across
school %>%
  filter(District %in% districtInOxfordshire$District) %>%
  group_by(Year, District) %>%
  mutate(ads = mean(`Attainment 8 score`, na.rm = TRUE)) %>%
  filter(row_number() == 1) %>%
  ggplot(aes(x = factor(District), y = ads, fill = factor(District))) +
  geom_bar(stat = "identity") +
  scale_fill_brewer(palette = "Paired") +
  labs(title = "Average Attainment 8 score In Oxfordshire", x = "District", y = "Average Attainment 8 score") +
  facet_wrap(~Year, ncol = 2, scales = "free_x") +
  theme_minimal() +
  theme(plot.title = element_text(hjust = 0.5),
        axis.text = element_text(color = "black", size = 8),
        axis.title = element_text(color = "black", size = 10),
        legend.position = "none")
```

Figure 102

Boxplot of average attainment 8 scores in Oxfordshire



This chart shows the side-by-side bar graph comparing the average attainment of 8 cores of Oxfordshire in two academic years. Like the chart of Yorkshire here also we see very little change in the average attainment 8 across districts. One change that is the most significant is that the average attainment of 8 scores of districts Vale of the white horse and Oxford both have around 10 score increase in the score over one academic year.

Crime

Figure 103

Read clean crime data.

```
crime<- read_csv("clean data\\crime.csv")
```

Figure 104

Sampling crime data.

```
#randomly selecting 5 district from each country
oxfordshire_districts<-crime %>%
  filter(`Falls within`=="Oxfordshire Police"&(year == 2022 | year==2021)&`Crime type`=="Drugs") %>%
  distinct(District) %>%
  sample_n(size = 5, replace = FALSE) %>%
  select(District)

yorkshire_districts<- crime %>%
  filter(`Falls within`=="Yorkshire Police"&(year == 2022 | year==2021)&`Crime type`=="Drugs") %>%
  distinct(District) %>%
  sample_n(size = 5, replace = FALSE) %>%
  select(District)
```

Randomly selecting 5 districts from each country for visualization.

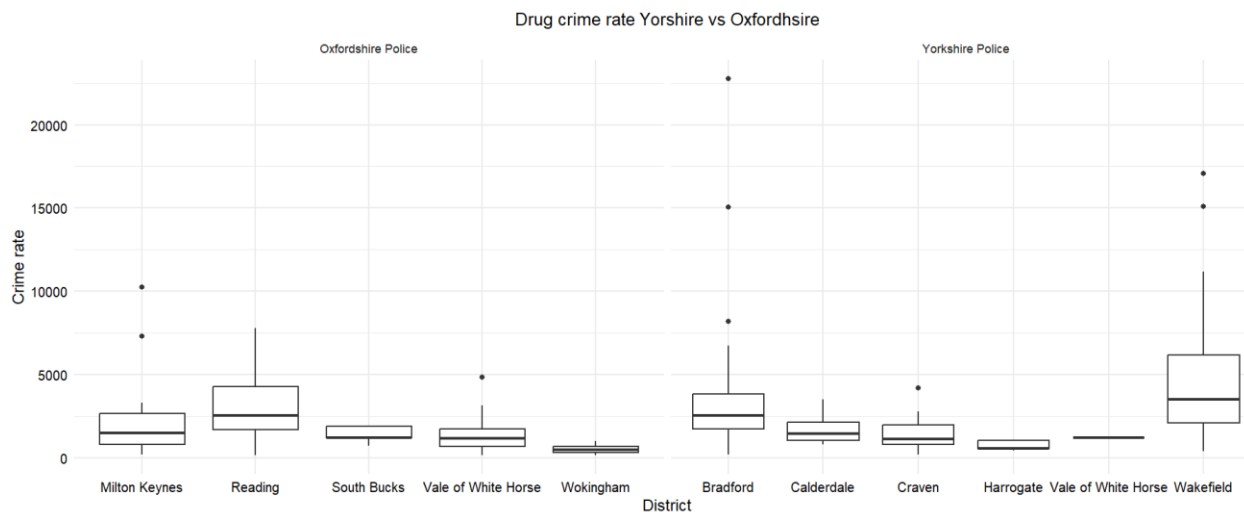
Boxplot showing Drug crime rate of Districts in Oxfordshire and Yorkshire.

Figure 105

Code to create a boxplot of drug crime.

```
#draw box plot
crime %>%
  filter(District %in%yorkshire_districts$District | District %in%oxfordshire_districts$District ) %>%
  # sample_n(size=100000) %>%
  group_by(LSOA) %>%
  mutate(COUNT = n()) %>%
  ggplot(aes(x = District, y = COUNT)) +
  geom_boxplot() +
  facet_wrap(~ `Falls within`, scales = "free_x") +
  labs(title = "Drug crime rate Yorshire vs Oxfordhsire", x = "District", y = "Crime rate") +
  theme_minimal()+
  theme(plot.title = element_text(hjust = 0.5),
        axis.text = element_text(color = "black", size = 10),
        axis.title = element_text(color = "black", size = 12),
        legend.title = element_blank(),
        legend.text = element_text(color = "black", size = 10))
```

In the code above select 5 districts from each country randomly and plot them in a grouped box plot side by side using facet wrap.

Figure 106*Box plot of drug crime*

In the above box plot, the median drug crime rates in Oxford and Yorkshire are the same but there are extreme outliers in the Yorkshire box plot where it shows that there are higher drug crime rates than normal. So, all over we can conclude that some places in the district in Yorkshire have a high drug crime rate, so the mean drug crime rate is higher in Yorkshire even if the median is similar in both countries.

Line graph showing Drug crime rate per thousand people from the year 2021 to 2023.

Figure 107

Code to create a line graph of drug crime rate.

```
#to calculate drug offence rate according to population
population<-read_csv("clean data\\populationData.csv")

#sum of population of district using shortpostcode
population_<- population %>%
  group_by(shortPostcode) %>%
  summarise(shortPostcode_population=sum(Population))

#create line graph
crime %>%
  filter(`Crime type` == "Drugs") %>%
  inner_join(population_, by = "shortPostcode") %>%
  mutate(date = parse_date_time(year, orders = "%y"), year=year(date)) %>%
  group_by(Country, year) %>%
  summarize(drug_rate_per_thousand = n() / sum(shortPostcode_population) * 1000, .groups = "drop") %>%
  ggplot(aes(x = year, y = drug_rate_per_thousand, color = Country)) +
  geom_line() +
  labs(x = "Year", y = "Drug Offense Rate per 1000 People", title = "Drug Offense Rate by Country") +
  theme_minimal()
```

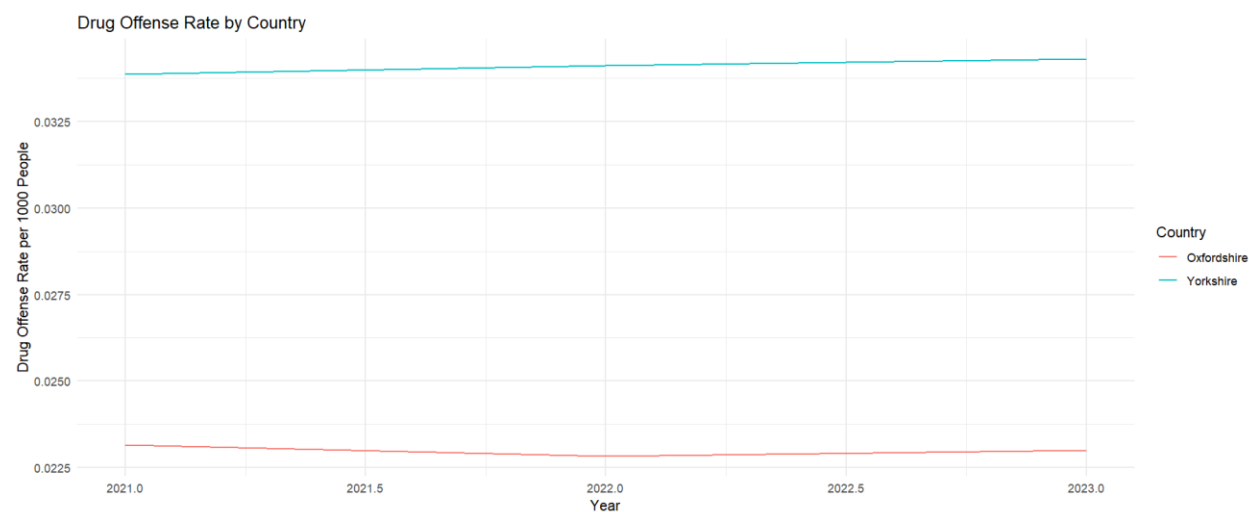
Figure 108

Code to create a line graph of drug crime rate.

```
#to calculate drug offence rate according to population
population<-read_csv("clean data\\populationData.csv")

#sum of population of district using shortpostcode
population_<- population %>%
  group_by(shortPostcode) %>%
  summarise(shortPostcode_population=sum(Population))

#create line graph
crime %>%
  filter(`Crime type` == "Drugs") %>%
  inner_join(population_, by = "shortPostcode") %>%
  mutate(date = parse_date_time(year, orders = "%y"), year=year(date)) %>%
  group_by(Country, year) %>%
  summarize(drug_rate_per_thousand = n() / sum(shortPostcode_population) * 1000) %>%
  ggplot(aes(x = year, y = drug_rate_per_thousand, color = Country)) +
  geom_line() +
  labs(x = "Year", y = "Drug Offense Rate per 1000 People", title = "Drug Offense Rate by Country") +
  theme_minimal()
```

Figure 109*Line graph of drug crime rate*

Overall, the crime rates in Yorkshire are way higher than in Oxfordshire but there is little change in the rates over the year. Except in the year 2022 in Oxfordshire, the drug crime rate has decreased by little and in Yorkshire in the same year the drug crime rate has increased by little.

Figure 110*Sampling robbery rate data*

```
#randomly selecting 5 district from each country
oxfordshire_districts_robbery<-crime %>%
  filter(`Falls within`=="Oxfordshire Police"&(year == 2022 | year==2021)&`Crime type`=="Robbery") %>%
  distinct(District) %>%
  sample_n(size = 5, replace = FALSE) %>%
  select(District)

yorkshire_districts_robbery<- crime %>%
  filter(`Falls within`=="Yorkshire Police"&(year == 2022 | year==2021)&`Crime type`=="Robbery") %>%
  distinct(District) %>%
  sample_n(size = 5, replace = FALSE) %>%
  select(District)
```

Pie chart showing robbery rate in Districts in Yorkshire and Oxfordshire

Figure 111

Code to create a pie chart of the robbery rate.

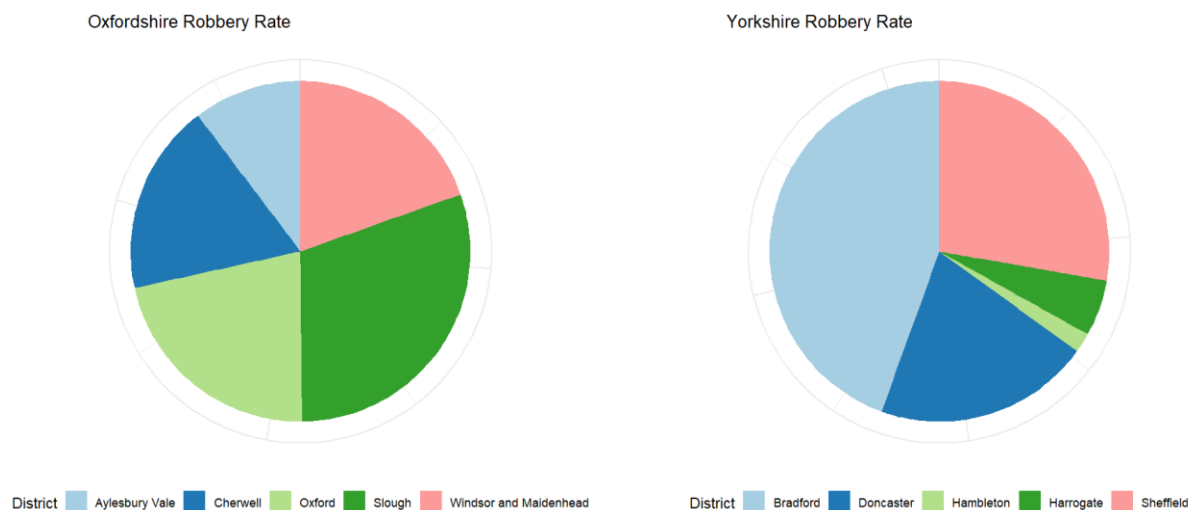
```
#draw pie chart
oxfordshire_pie<- crime %>% filter(District %in% oxfordshire_districts_robbery$District) %>%
  group_by(District) %>%
  summarise(count=n()) %>%
  ggplot( aes(x = "", y = count, fill = District)) +
  geom_bar(width = 1, stat = "identity") +
  coord_polar("y", start = 0) +
  labs(title = "Oxfordshire Robbery Rate", fill = "District")+
  theme_minimal() +
  theme(legend.position = "bottom",
        axis.text = element_blank(),
        axis.title = element_blank())+
  scale_fill_brewer(palette = "Paired")

#draw pie chart
yorkshire_pie<-crime %>% filter(District %in% yorkshire_districts_robbery$District) %>%
  group_by(District) %>%
  summarise(count=n()) %>%
  ggplot( aes(x = "", y = count, fill = District)) +
  geom_bar(width = 1, stat = "identity") +
  coord_polar("y", start = 0) +
  labs(title = "Yorkshire Robbery Rate", fill = "District")+
  theme_minimal() +
  theme(legend.position = "bottom",
        axis.text = element_blank(),
        axis.title = element_blank())+
  scale_fill_brewer(palette = "Paired")

# Display the radar charts side by side
grid.arrange(oxfordshire_pie, yorkshire_pie, ncol = 2)
```

The code above randomly selects 5 districts from the countries Oxfordshire and Yorkshire each then creates two separate pie charts showing the robbery rate in the selected districts.

Lastly, these two plots are displayed side by side using a grid arrangement.

Figure 112*Pie chart of robbery rate.*

In the pie chart above we can see that most districts in Oxfordshire have a somewhat similar robbery rate except Slough which has a higher rate. But in Yorkshire Bradford covers almost half of the robbery rate from the randomly selected districts. Districts Doncaster and Sheffield have similar rates, that is the second highest of Yorkshire's sample districts then Harrogate and Hambleton with the smallest robbery rate.

Radar chart showing vehicle crime Rate in Districts in Yorkshire and Oxfordshire

Figure 113

Code to create radar chart of vehicle crime rate.

```
#random 3 district to show
oxfordshire_vehical_crime<- crime %>%
  filter(`Crime type` == "Vehicle crime" & Country=="Oxfordshire") %>%
  distinct(District) %>%
  sample_n(size = 3, replace = FALSE) %>%
  select(District)

#prepare data to create radar chart
data_oxforshisre <- crime %>%
  filter(District %in% oxfordshire_vehical_crime$District ) %>%
  group_by(District, year) %>%
  mutate(count=n()) %>%
  summarise(count=n()) %>%
  pivot_wider(names_from = year, values_from = count)

# Create the radar chart oxfordshire
radar_plot_oxfordshire <- data_oxforshisre %>%
  ggradar(
    grid.label.size = 0,
    axis.label.size = 5,
    group.point.size = 3,
    grid.max = 150000 # Adjust the maximum value for the grid
  )+
  labs(title = "Oxfordshire Radar chat")+
  theme(plot.title = element_text(size=18,hjust = 0.5),
        legend.text = element_text(color = "black", size = 10))
```

Figure 114

Code to create radar chart of vehicle crime rate.

```
#select random 3 district to show
yorkshire_vehical_crime<- crime %>%
  filter(`Crime type` == "Vehicle crime" & Country=="Yorkshire") %>%
  distinct(District) %>%
  sample_n(size = 3, replace = FALSE) %>%
  select(District)

#prepare data to show in radar chart
data_yorkshire <- crime %>%
  filter(`Crime type` == "Vehicle crime"& District %in% yorkshire_vehical_crime$District) %>%
  group_by(District, year) %>%
  summarise(count = n()) %>%
  pivot_wider(names_from = year, values_from = count)

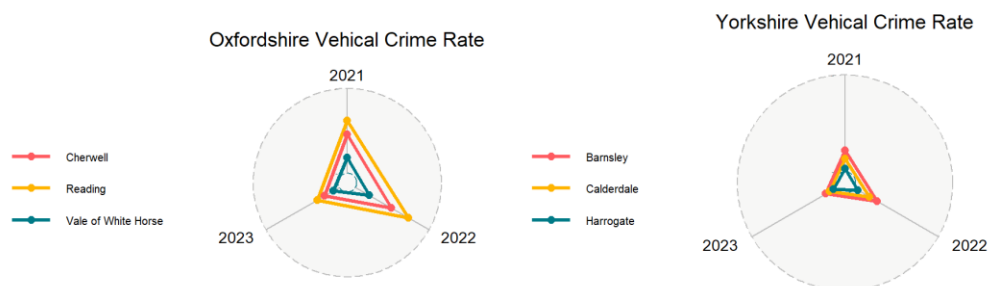
# Create the radar chart yorkshire
radar_plot_yorkshire <- data_yorkshire %>%
  ggradar(
    grid.label.size = 0,
    axis.label.size = 5,
    group.point.size = 3,
    grid.max = 35000 # Adjust the maximum value for the grid
  )+
  labs(title = "Yorkshire Radar chat")+
  theme(plot.title = element_text(size=18,hjust = 0.5),
        legend.text = element_text(color = "black", size = 10))

# Display the radar charts side by side
grid.arrange(radar_plot_oxfordshire, radar_plot_yorkshire, ncol = 2)
```

In the code above select 3 districts randomly from each country to be shown in the chart which is filtered to keep data of vehicle crime only then format the data to create a radar chart. Create 2 radar charts each for one country and then both charts are shown side by side using grid arrangement.

Figure 115

Radar chart of vehicle crime rate



In the above radar chart, there is a common trend in vehicle crime as the crime rate is significantly higher in the year 2022 in all districts in both countries. Moreover, in both countries the change in vehicle crime rate is the similar over the year in all district.

Linear Modeling

House price and Average download speed

Figure 116

Code to create a linear model of house price and download speed.

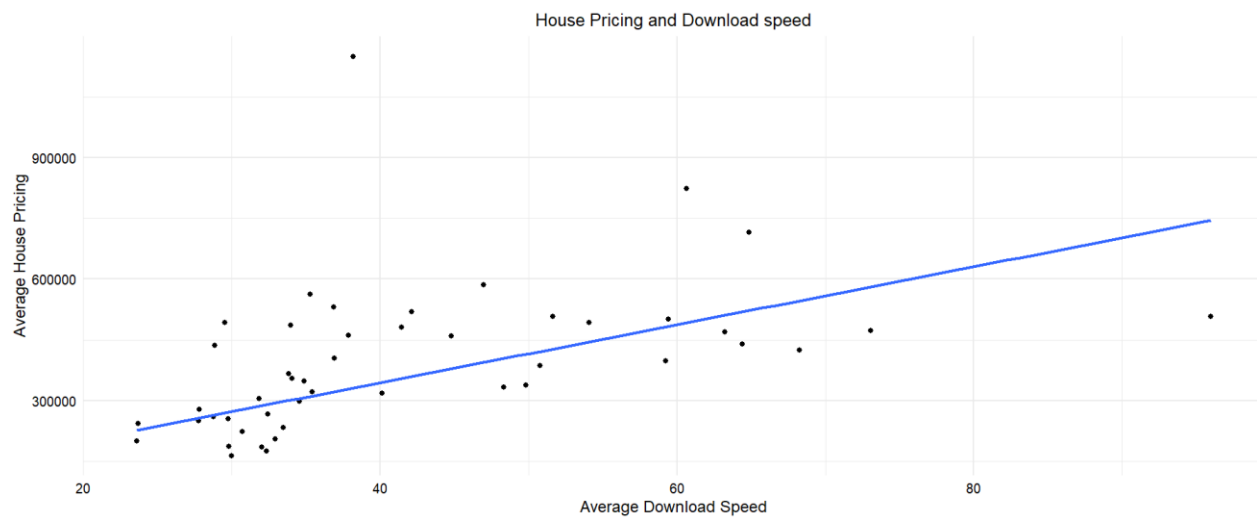
```
housePricing <- read_csv("clean_data\\housingData.csv")
internet<- read_csv("clean_data\\internetData.csv")

housePricing %>%
  inner_join(internet,by="Postcode") %>%
  group_by(shortPostcode.x) %>%
  mutate(ads=mean(`Average download speed (Mbit/s)`),average_price=mean(Price)) %>%
  ggplot(aes(x=ads,y=average_price)) +
  geom_point() +
  geom_smooth(method="lm",se=FALSE) +
  scale_fill_brewer("PuOr")+
  labs(title = "House Pricing and Download speed",x="Average Download Speed", y= "Average House Pricing") +
  theme_minimal() +
  theme(plot.title = element_text(hjust = 0.5),
        axis.text = element_text(color = "black", size = 10),
        axis.title = element_text(color = "black", size = 12),
        legend.title = element_blank(),
        legend.text = element_text(color = "black", size = 10))
```

Here, join data of internet and house pricing using postcode column then calculate average house pricing and download speed of each district which is plotted as a scatter plot then a line of best fit is drawn.

Figure 117

Linear model of house price and download speed.



The chart shows that there is a statistically significant positive relationship between Average House Pricing and average download speed as the points are scattered around the line. This means that with the increase in download speed, house pricing will also increase except in some outlier cases.

House pricing and drug crime rate per thousand

Figure 118

Code to create a linear model of house price and drug crime rate.

```
housePricing <- read_csv("clean data\\housingData.csv")
crime<- read_csv("clean data\\crime.csv")

#to calculate drug crime rate per thousand
population<- read_csv("clean data//populationData.csv")

#sum of population of district using shortpostcode
population_<- population %>%
  group_by(shortPostcode) %>%
  summarise(shortPostcode_population=sum(Population))

#crime rate per 1000 people for each district
crime_<-crime %>%
  filter(`Crime type`=="Drugs") %>%
  inner_join(population_,by="shortPostcode") %>%
  group_by(shortPostcode) %>%
  mutate(count=n()) %>%
  filter(row_number() == 1) %>%
  mutate(drug_crime_rate=(count/shortPostcode_population)*1000)

#calculate average housepricing of each district
housePricing_<-housePricing %>%
  group_by(shortPostcode) %>%
  mutate(mean_price_district=mean(Price)) %>%
  slice(1) %>%
  select(mean_price_district)
```

Figure 119

Code to create a linear model of house price and drug crime rate.

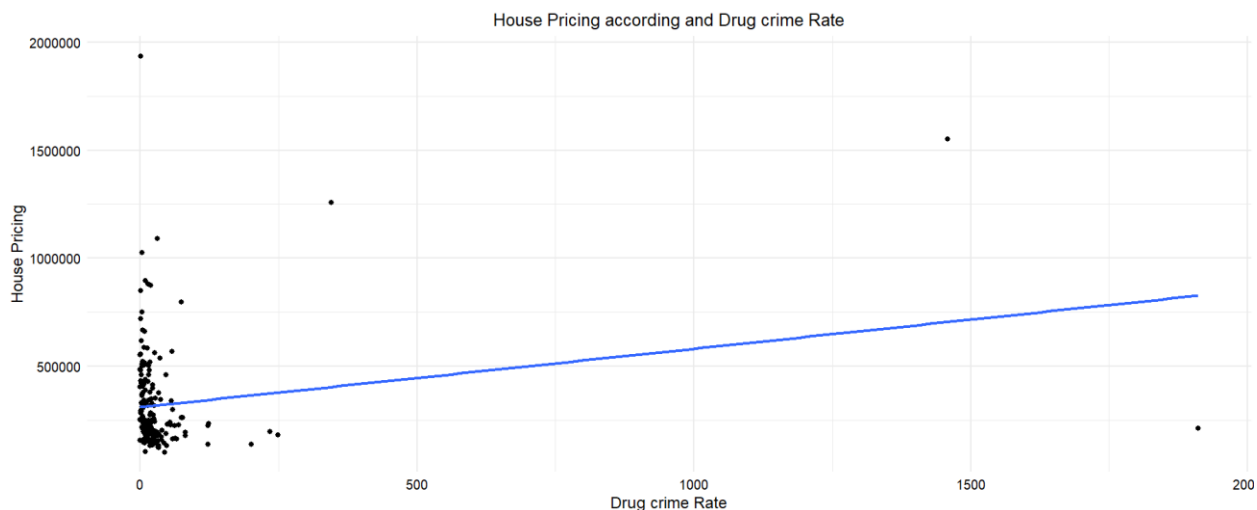
```
#join housepricing and crime data
housepricing_crime_ <- housePricing_ %>%
  inner_join(crime_, by="shortPostcode")

#create linear model
housepricing_crime_ %>%
  ggplot(aes(x=drug_crime_rate,y=mean_price_district)) +
  geom_point() +
  geom_smooth(method="lm",se=FALSE) +
  scale_fill_brewer("PuOr")+
  labs(title = "House Pricing according and Drug crime Rate",x="Drug crime Rate", y= "House Pricing") +
  theme_minimal() +
  theme(plot.title = element_text(hjust = 0.5),
        axis.text = element_text(color = "black", size = 10),
        axis.title = element_text(color = "black", size = 12),
        legend.title = element_blank(),
        legend.text = element_text(color = "black", size = 10))
```

The above-calculated drug crime rate per thousand population for each district and then joined with the data set providing information on average house pricing in each district. The newly joined dataset creates a linear model showing the change in House pricing according to the crime rate.

Figure 120

A linear model of house price and drug crime rate



Here we can see that in most of the districts, the crime rate is low other than some extreme outliers. Here the regression line shows a positive relationship between drug crime rate and house pricing. But this is not a strong relationship, rather it could be because of the extreme outliers that we can see on the chart.

Average attainment 8 score and house price

Figure 121

Code to create a linear model of average attainment 8 scores and house price.

```
school <- read_csv("clean data\\schoolData.csv")
housePricing <- read_csv("clean data\\housingData.csv")

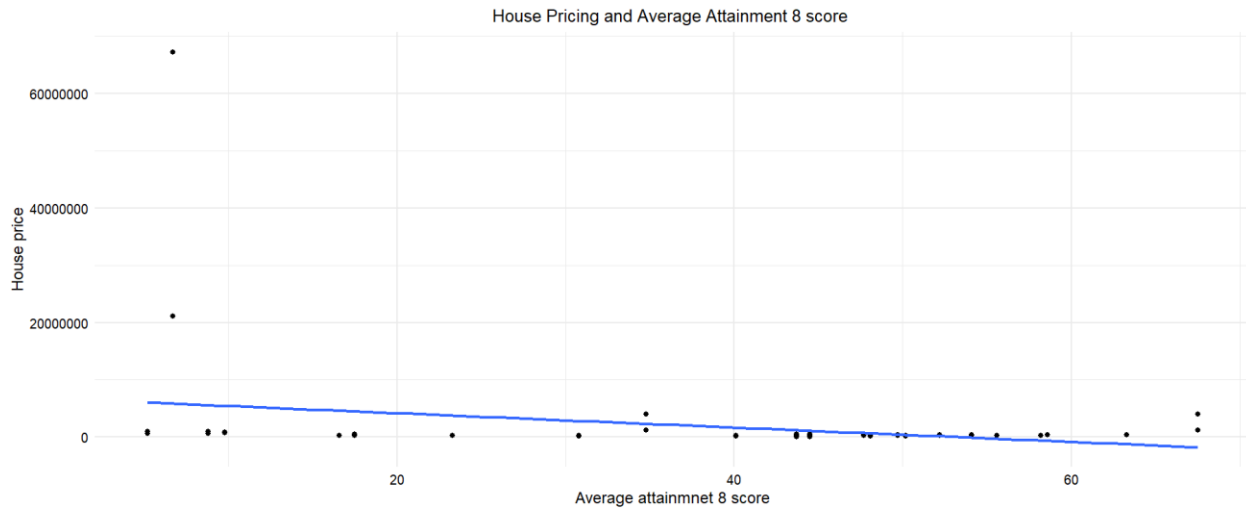
school_housePricing<-school %>%
  inner_join(housePricing, by="Postcode")

school_housePricing %>%
  ggplot(aes(x=`Attainment 8 score`,y=Price)) +
  geom_point() +
  geom_smooth(method="lm",se=FALSE) +
  scale_fill_brewer("PuOr")+
  labs(title = "House Pricing and Average Attainment 8 score ",x="Average attainmnet 8 score", y= "House price") +
  theme_minimal() +
  theme(plot.title = element_text(hjust = 0.5),
        axis.text = element_text(color = "black", size = 10),
        axis.title = element_text(color = "black", size = 12),
        legend.title = element_blank(),
        legend.text = element_text(color = "black", size = 10))
```

Here we take the value from school and housing data where there are common postcode columns. This is then used to create a scatter plot and a regression line is drawn over it.

Figure 122

A linear model of average attainment 8 scores and house price



Here the above plot shows that the house pricing decreases with the increase in the attainment 8 score. The line is slightly sloped so the effect of average attainment on house pricing is a little decrease in house pricing with the increase in average attainment 8 score. Other than that, you can see that there are some outliers where house pricing is high when the average attainment 8 score is on the lower side.

Average download speed and Drug crime rate

Figure 123

Code to create a linear model of download speed and drug crime rate.

```
internet<- read_csv("clean data\\internetData.csv")
crime<- read_csv("clean data\\crime.csv")

#to calculate drug crime rate per thousand
population<- read_csv("clean data//populationData.csv")

#sum of population of district using shortpostcode
population_<- population %>%
  group_by(shortPostcode) %>%
  summarise(shortPostcode_population=sum(Population))

#crime rate per 1000 people for each district
crime_<-crime %>%
  filter(`Crime type`=="Drugs") %>%
  inner_join(population_,by="shortPostcode") %>%
  group_by(shortPostcode) %>%
  mutate(count=n()) %>%
  filter(row_number() == 1) %>%
  mutate(drug_crime_rate=(count/shortPostcode_population)*1000)

#calculate average
internet_<- internet %>%
  group_by(shortPostcode) %>%
  mutate(avg_download_speed= mean(`Average download speed (Mbit/s)`, na.rm=TRUE)) %>%
  slice(1) %>%
  select(avg_download_speed)
```

Figure 124

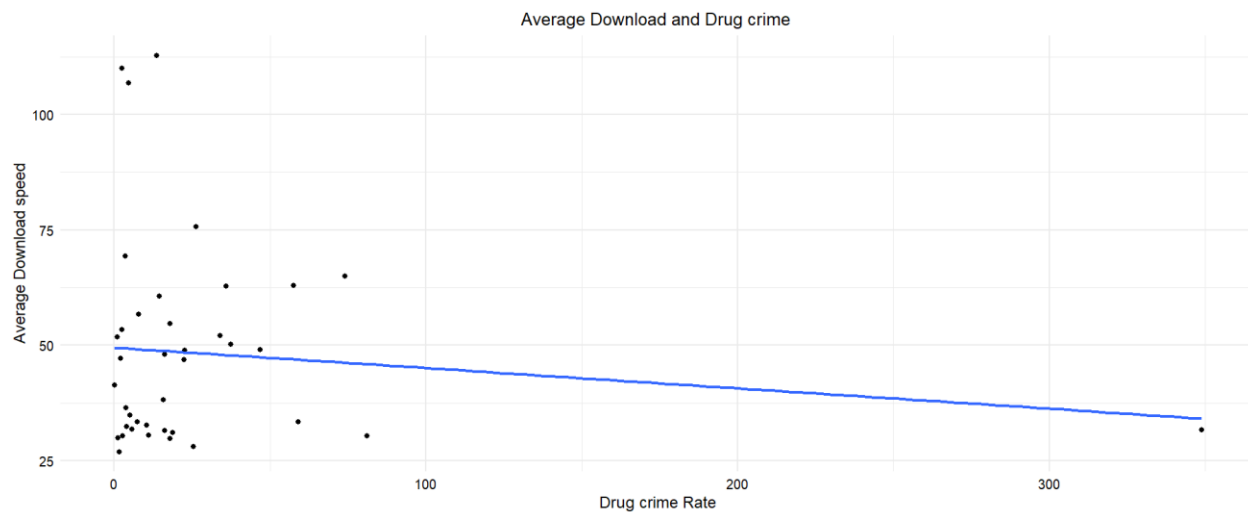
Code to create a linear model of download speed and drug crime rate.

```
crime_internet_ <- crime_ %>%
  inner_join(internet_, by = "shortPostcode")

crime_internet_ %>%
  ggplot(aes(x=drug_crime_rate,y=avg_download_speed)) +
  geom_point() +
  geom_smooth(method="lm",se=FALSE) +
  scale_fill_brewer("PuOr")+
  labs(title = "Average Download and Drug crime ",x="Drug crime Rate", y= "Average Download speed") +
  theme_minimal() +
  theme(plot.title = element_text(hjust = 0.5),
        axis.text = element_text(color = "black", size = 10),
        axis.title = element_text(color = "black", size = 12),
        legend.title = element_blank(),
        legend.text = element_text(color = "black", size = 10))
```

Figure 125

A linear model of download speed and drug crime rate.



In the above linear model, the data points are very scattered which can be concluded to have no relation between drug crime rate and average download speed.

Average attainment 8 score and average download speed

Figure 126

Code to create a linear model of average attainment 8 scores and download speed.

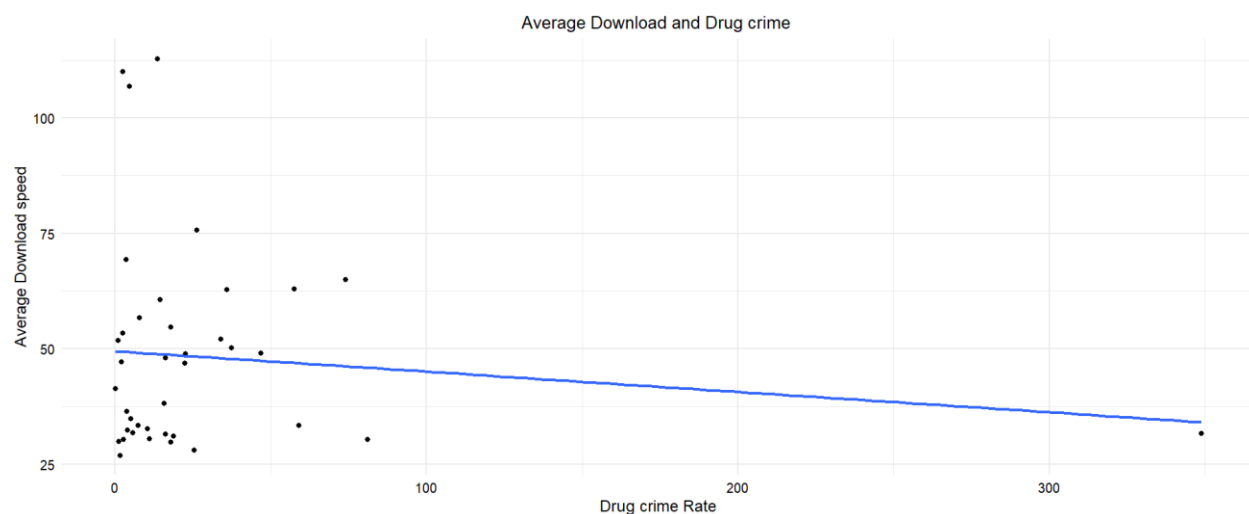
```
school <- read_csv("clean data\\schoolData.csv")
internet<- read_csv("clean data\\internetData.csv")

school_internet_<-school %>%
  inner_join(internet,by="Postcode")

school_internet_ %>%
  ggplot(aes(x=`Average download speed (Mbit/s)`,y=`Attainment 8 score`)) +
  geom_point() +
  geom_smooth(method="lm",se=FALSE) +
  scale_fill_brewer("PuOr")+
  labs(title = "Average Attainment rate vs Average download speed ",x="Average download speed", y= "Attainment 8 rate") +
  theme_minimal() +
  theme(plot.title = element_text(hjust = 0.5),
        axis.text = element_text(color = "black", size = 10),
        axis.title = element_text(color = "black", size = 12),
        legend.title = element_blank(),
        legend.text = element_text(color = "black", size = 10))
```

Figure 127

Linear model of average attainment 8 scores and download speed.



Looking at the liner regressing plot above the data points don't follow the regressing line. This means that the relationship between drug crime rate and average download speed is not statistically significant.

Recommendations

The code below provides recommendations for friends who are moving in for further studying by considering house price, internet, crime, and school data.

Figure 128

Read clean data for Recommendation.

```
#read all clean data
housePricing <- read_csv("clean data\\housingData.csv")
internet<-read_csv("clean data\\internetData.csv")
crime<-read_csv("clean data\\crime.csv")
school<-read_csv("clean data\\schoolData.csv")
```

Figure 129

Calculate the average house price across the district.

```
#calculate average house pricing accorsing to district
housePricing<-housePricing %>%
  group_by(District) %>%
  mutate(average_housePrice=mean(Price)) %>%
  filter(row_number() == 1)
```

Figure 130

Calculate the average internet speed across the district.

```
#calculate average internet speed using upload and download speed
internet <- internet %>%
  group_by(District) %>%
  mutate(average_internet_speed = (mean(`Average download speed (Mbit/s)`, na.rm = TRUE)+
                                   mean(`Average upload speed (Mbit/s)`, na.rm = TRUE))/2) %>%
  filter(row_number() == 1)
```

Figure 131

Calculate the average crime rate across the district.

```
#to calculate crime rate
population<- read_csv("clean data//populationData.csv")

#Calcuate crime rate per thousand population in the district
#sum of population of district using shortpostcode
population_<- population %>%
  group_by(shortPostcode) %>%
  summarise(shortPostcode_population=sum(Population))

#crime rate per 1000 people for each district
crime<-crime %>%
  inner_join(population_,by="shortPostcode") %>%
  group_by(District) %>%
  mutate(count=n()) %>%
  filter(row_number() == 1) %>%
  mutate(crime_rate=(count/shortPostcode_population)*1000)
```

Figure 132

Calculate the average attainment 8 score across the district.

```
# Calculate the mean of ATT8SCR by District
school <- school %>%
  group_by(District) %>%
  mutate(average_attatinmnet8Score = mean(`Attainment 8 score` , na.rm = TRUE)) %>%
  filter(row_number() == 1)
```

Figure 133

Combine all calculated data.

```
#combine all data
all<-housePricing %>%
  inner_join(internet,by="District") %>%
  inner_join(crime,by="District") %>%
  inner_join(school,by="District") %>%
  select(Country=Country.x,Postcode=Postcode.x,District,average_housePrice,average_attatinmnet8Score,average_internet_speed,crime_rate)
```

In the above code, firstly read clean data proving information on attainment 8 score, internet, crime rate and house pricing. Then calculate the average of all the variables by grouping them by districts. For internet data average internet speed is calculated by taking the average of upload and download speed. After calculating the data by district join the data by district. Here as inner join is done the rows with all values are only taken into consideration.

Figure 134

Checking for extreme outliers.

```
#checking for extreme values(outliers)

# Calculate IQR for each variable
iqr_housePrice <- IQR(all$average_housePrice)
iqr_attainment8Score <- IQR(all$average_attatinmnet8Score)
iqr_internet_speed <- IQR(all$average_internet_speed)
iqr_crime_rate <- IQR(all$crime_rate)

# Calculate lower and upper bounds
upper_bound_housePrice <- quantile(all$average_housePrice, 0.75) + 1.5 * iqr_housePrice

lower_bound_attainment8Score <- quantile(all$average_attatinmnet8Score, 0.25) - 1.5 * iqr_attainment8Score

lower_bound_internet_speed <- quantile(all$average_internet_speed, 0.25) - 1.5 * iqr_internet_speed

upper_bound_crime_rate <- quantile(all$crime_rate, 0.75) + 1.5 * iqr_crime_rate

# Identify potential outliers
outliers_housePrice <- all$average_housePrice > upper_bound_housePrice
outliers_attainment8Score <- all$average_attatinmnet8Score < lower_bound_attainment8Score
outliers_internet_speed <- all$average_internet_speed < lower_bound_internet_speed
outliers_crime_rate <- all$crime_rate > upper_bound_crime_rate

#filter the outliers
outliers <- data.frame(
  average_housePrice = all$average_housePrice[outliers_housePrice],
  average_attatinmnet8Score = all$average_attatinmnet8Score[outliers_attainment8Score],
  average_internet_speed = all$average_internet_speed[outliers_internet_speed],
  crime_rate = all$crime_rate[outliers_crime_rate]
)
```

Here, the code finds potential outliers which are extreme values that can influence the suitability of a district for students to live in. For example, high house prices might make a district less affordable for students, while low school attainment scores might indicate lower-quality education options. On the other hand, areas with very high crime rates might be less safe for students.

These values can be used in the recommendation algorithm below for example a district may have cheap house pricing but a very high crime rate. In such cases we still determine such places to be unsuitable for stay despite being affordable due to safety concerns.

Figure Error! Unknown switch argument.

Outliers check result.

```
> # Print out the outliers
> print(outliers)
[1] average_housePrice      average_attatinmnet8Score average_internet_speed  crime_rate
<0 rows> (or 0-length row.names)
```

Above we can see that there are no extreme values that are required to be removed.

Figure 136

Taking the top 5 recommended places.

```
#get top 5 district for recommendation
all %>% mutate(allover=(average_attatinmnet8Score+average_internet_speed-average_housePrice-crime_rate)^2) %>%
  arrange(desc(allover)) %>%
  top_n(5)
```

Finally, when calculating the overall score for a recommendation. We add the score that should be higher including average download speed and average attainment 8 score then subtract scores that should be lower including average house pricing and crime rate. Then by arranging the data by allover column in descending order, we get the best places for buying housing within Oxfordshire and Yorkshire. Additionally, in the results, you can see the average housing pricing, attainment 8 score, internet speed and crime rate too.

Figure 137

Top 5 recommended places.

	Country	Postcode	District	average_housePrice	average_attatinmnet8Score	average_internet_speed	crime_rate	allover
1	OXFORDSHIRE	OX52NP	Cherwell	400374.2	50.2	27.78237	12676.527	170546459324
2	YORKSHIRE	DL78JL	Hambleton	319020.0	58.2	19.99744	1705.601	102814760610
3	YORKSHIRE	HG50AX	Harrogate	406131.8	44.0	18.75905	3054.178	167381814719
4	OXFORDSHIRE	OX43JF	Oxford	682980.0	2.7	34.52015	8016.157	477424314979
5	YORKSHIRE	YO607BW	Ryedale	328232.6	52.6	18.45029	1081.598	108401050793

The results show that Cherwell in Oxfordshire is the best place to buy property for a student. Other than that, other top 5 recommendations include Hambleton in Yorkshire, Harrogate in Yorkshire, Oxford in Oxfordshire, and Ryedale in Yorkshire. These places are recommended based on high internet speed and attainment 8 score and lower crime rate and house pricing.

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Appendix

One drive code and data set

Data science Assignment